PAQ: Prefill-Once, Ask-Many

Reversible, query-conditioned below-page KV selection for multi-query document VQA

A living research document — current state, honest negatives, and roadmap

(status: 2026-07-17; this document tracks ongoing work and is updated as results land)

BASED-MM program

July 17, 2026

Abstract

Real document-VQA benchmarks are *multi-query*: one long document is prefilled once and asked many different questions (in our pinned evaluation sets, 5.5–5.8 gold-bearing questions per document). Standard KV-cache eviction commits to *one* compressed context and reuses it for every question. We study PAQ ("Prefill-once, Ask-many") and its below-page instantiation PAQ-KV: prefill the full document *once* into the KV cache of a frozen reader (Qwen3-VL-8B) and retain it; per query, re-select the most relevant 64-token, below-page KV blocks with the reader's own question→content attention; decode under a restricted-attention **4D mask over the retained full-context cache** — no re-read, no gathering (M-RoPE positions untouched), no second model, no index. Selection is fully **reversible**: the next question re-selects from the same intact cache (§4.5).

What is confirmed (all on the development pin; the sealed test set is untouched). **(1) Reversibility — the mechanistic contribution.** Per-query re-selection beats the identical scorer committed once, on both cells (confirmatory run, 903 queries, 80 documents/cell, document-clustered inference: +0.257/+0.163 per cell, LB_{95} +0.184/+0.126), and beats a strong gold-informed static subset (+0.108/+0.104, LB_{95} +0.045/+0.063). The same fresh-over-stale collapse holds for a retrieval scorer (ColQwen2 fresh 0.423 vs. stale 0.178 pooled). The contrast survived the fix of a KV-cache reuse bug under a clean isolation re-run (+0.264, LB_{95} = +0.187; §4.3) and *reappears at block level* (+0.379, LB_{95} = +0.273). **(2) A powered, audited beat on MMLongBench-Doc.** PAQ-KV at a nominal 5% decode-attention budget scores 0.404 vs. ColQwen2-fresh 0.325 at the matched 5% budget — +0.079, document-clustered LB_{95} = +0.041, 80 documents / 444 queries — while *matching* the gold-page oracle at that budget (−0.026, CI straddles zero; we do not claim to exceed it) and matching full context (+0.009, n.s.). The mechanism is audited: selection is query-*dependent* (+0.329 over random blocks, LB_{95} = +0.238), query-*specific* (+0.698 over wrong-question selection, LB_{95} = +0.603 — ruling out leakage and query-invariance), reversible, and below-page granularity beats page-level granularity (+0.054, LB_{95} = +0.011). **(3) An honest, characterized limitation on LongDocURL.** PAQ-KV matches but does not beat ColQwen2 there (+0.019, LB_{95} = −0.019, 80 documents / 463 queries). We can say why: at 80 documents the significance bar is a ≈ +0.039 accuracy delta; only ≈10% of queries are selection-capturable at all; ≈41% score zero even when the reader is handed the gold pages at native resolution (a reader-conversion wall, not a selection wall); and large documents are a low-leverage dead zone for every arm (§7.5.2).

Negatives, reported at the same prominence. The off-benchmark LoRA/per-head-mixture selection lane (B) is **stopped**: the 1,152-head mixture overfits its off-benchmark training distribution and transfers *worse* than a single externally selected head (pooled −0.023 [−0.037, −0.008]); its byproduct is that expert head L21.H18 transfers off-benchmark. Swapping that sharp head in as the PAQ-KV block scorer (C-V1) was a **no-go**: it does not separate from the diffuse mean-attention scorer on LongDocURL (LB_{95}(V1 − V0) = −0.011, n.s.) and *hurts*

MMLongBench-Doc (0.402 → 0.381) — recall-optimized selection is not accuracy-optimized selection. The page-level history stands in full: PAQ v1 lost to ColQwen2-fresh by ≈10 points (pooled −0.099), PAQ-T (expert-head) reached accuracy *parity* but not a beat, and CAPS (ACL 2026) published the page-level method.

Status. Every number here is a development-pin result (the 80-document/cell confirmatory pin, now development-only). The sealed fresh test set (73/45 documents) is **untouched**, reserved for a single one-touch final evaluation that has not run; the scope decision on the LongDocURL cell is held for a human fork (§9.3).

Contents

1 Introduction

1.1 The problem

Long-context vision–language models (VLMs) can read an entire multi-page document, but they pay for every page on every question. The standard efficiency response, inherited from single-query text benchmarks, is to *evict*: keep a compact subset of the KV cache (SnapKV, H2O, FastV, LOOK-M, KVzip) and decode against it (Li et al., 2024; Zhang et al., 2023; Chen et al., 2024; Wan et al., 2024; Kim et al., 2025). Eviction is a *single, irreversible commitment*: whatever subset is kept must serve every future question about that document.

1.2 The multi-query regime

Document assistants are not asked one question. The two long-document VQA benchmarks we use are natively multi-query: grouping the 64K splits by source document, MMLongBench-Doc (Ma et al., 2024) averages ≈ 6.97 questions per document and LongDocURL (Deng et al., 2025) ≈ 3.6; inside our pinned evaluation subsets (which require ≥ 3 gold-bearing questions per document) the averages are 5.50 and 5.79 respectively (Table 1). Different questions need different pages, so a subset committed for question 1 is *stale* for question 5.

1.3 The question this project asks

> *At a matched page budget on long multimodal documents, does per-query* ***re-selection*** *of context beat a* ***commit-once*** *selection — and can a training-free, reader-internal selector compete with a dedicated trained retriever?*

The answer, on the evidence in this document, is split, and we state it up front:

- **Yes** to reversibility: re-selection beats commit-once, robustly, on both cells, for our scorer *and* for a retrieval scorer (§7.1, §7.2), at page level *and* at block level (§7.5). This is the confirmed mechanistic contribution.

- **Yes, below page granularity, on one cell of two**: PAQ-KV — per-query reversible 64-token KV-block selection over one retained prefill, decoded under a restricted-attention mask — **beats** ColQwen2-fresh on MMLongBench-Doc at matched budget (+0.079, $\mathrm{LB}_{95} = +0.041$, 80 documents, audited for query-dependence, query-specificity, and leakage), while *matching* the gold-page oracle at a 5% budget (§7.5.1).

- **Not (yet) on LongDocURL — and we can say why**: PAQ-KV matches ColQwen2 there (+0.019, $\mathrm{LB}_{95} = -0.019$) and the shortfall is characterized rather than hand-waved: the cell is mostly reader-conversion-bound, not selection-bound (§7.5.2).

- **No, at page level** (retained in full as history): PAQ v1 loses to ColQwen2-fresh (Faysse et al., 2025) by $\approx$10 accuracy points (§7.2); CAPS (Zhu et al., 2026) published the page-level attention-selection method (§8.1); cross-validated expert-head scoring (PAQ-T) recovers the whole gap to **parity**, not past it (§7.4). This is what pushed the program below page granularity.

Table 1: The confirmatory evaluation pin (the numbers that should be cited). "Complete queries" are rows where every gated arm scored (complete-case pairing); 4 of 907 vision queries are missing due to a per-document timeout (§7.1).

	LongDocURL	MMLongBench-Doc
Documents	80	80
Complete queries at gate (b=5%)	463	440
Gold-bearing queries / document (pin)	5.79	5.50
Median (mean) union pages / document	44 (45.0)	28 (36.5)
Benchmark-level questions / document (64K split)	≈ 3.6	≈ 6.97

1.4 Reading guide

§2 fixes benchmarks and pins; §3 the executed baseline panel; §4 the methods — including a methods-integrity note on a KV-cache bug found and fixed mid-program (§4.3), the executed PAQ-T accuracy read-eval protocol (§4.4), the PAQ-KV method in depth (§4.5, with Figures 2 and 3), and the executed-and-stopped B lane (§4.6); §5 the statistical machinery; §6 the algorithms; §7 all results (positive and negative, with per-cell results primary and pooled results explicitly labelled) — the PAQ-KV results at power are §7.5 and the lane negatives §7.6; §8 related work including the CAPS overlap; §9 current state, the held scope fork, and roadmap; §10 limitations. The appendix carries the full panel, the adversarial review trail, and reproducibility pins.

2 Benchmarks and setup

2.1 Cells, documents, pins

We evaluate on two cells: **LongDocURL** (Deng et al., 2025) and **MMLongBench-Doc** (Ma et al., 2024), both consumed as page images (no OCR). For each document we build the *union of the benchmark's own-document pages* across that document's queries (dropping cross-document 64K padding), so every query's evidence is present in one prefill. Documents qualify if they have ≥ 3 gold-bearing queries and ≤ 110 union pages ($\approx$64K tokens at read resolution); selection is seed-0 frozen.

Two pins exist: the 30-document/cell *screen* pin and the 80-document/cell *confirmatory* pin. The screen pin is a strict prefix of the confirmatory pin (same seed, same arrival order), which is why §7.1 additionally reports an *independent-extension* analysis on the 50 *new* documents per cell — the confirmatory is not independent of the screen unless restricted to those. Both pins are sha-256–checksummed and the harness verifies the screen pin byte-identically before use (Appendix C).

2.2 Reader, budgets, metric

The reader is **Qwen3-VL-8B** (Qwen Team, 2025), frozen, eager attention. All selection arms operate at a matched *page* budget $b \in \{2.5\%, 5\%, 10\%\}$ of the document's pages (the pre-registered gate budget is $b = 5\%$); the selected pages are re-rendered at full resolution and read. Accuracy is the benchmark's document-VQA scorer (exact/relaxed match by answer format). Scope note: everything in this document is a **Qwen3-VL-8B case study**; no second VLM has been run (§10).

Table 2: Arm taxonomy. The load-bearing axis is **reversible** (re-select per query) vs. **commit-once** (one subset per document, reused for all its queries); the second axis is deployability. Disclosed proxies are marked.

Arm	Reversible?	Deployable?	Role
PAQ (ours)	yes	yes	the method
attn_max (ours, max-agg)	yes	yes	reported alternate aggregation
paq_stale	no	yes	**isolation**: same scorer, committed once
gold_frequency_static	no	no (uses gold)	**isolation**: strong gold-informed static
ColQwen2-fresh	yes	yes	**strong deployable retrieval**, re-ranked per query
ColQwen2-stale	no	yes	retrieval committed once (same commit policy)
CLIP-fresh	yes	yes	cheaper retrieval control (Radford et al., 2021)
SnapKV-fresh	yes	yes	reference (weak port; beating it not required)
SnapKV-stale	no	yes	strongest stale family arm
H2O / FastV / LOOK-M / KVzip	no	yes	demoted query-blind controls (proxies, see text)
random / truncation	no	—	controls
full-context / oracle-evidence	—	—	reference rows, *not* competitors

2.3 The base-model ceiling

A large fraction of queries receive no credit from this reader even when it is given *exactly* the annotated gold-evidence pages: on the confirmatory run the oracle-evidence arm scores 0 on $317/903 = 35.1\%$ of queries and scores a full 1 on only $407/903 = 45.1\%$ (the pre-specified "answerable subset"). This limits absolute scores. Pairing controls for shared query difficulty, but does not imply that the ceiling affects every selector equally; we report the answerable subset only as a secondary view because it conditions on a model outcome. The oracle row is *not* a strict ceiling: PAQ beats oracle-evidence on 28 queries, because a selected page set can differ from the annotated gold set in ways that help the reader.

3 Baselines: the executed panel

Every arm below was actually run, per query, at matched page budget, on the same pinned data with the same reader. The resulting panel directly compares retrieval and reader-internal selection on MMLongBench-Doc/LongDocURL at 64K in the multi-query regime; our current literature search did not identify an earlier executed panel with this exact scope. Table 2 gives the taxonomy; Appendix A the full numbers.

The demoted eviction family (disclosed proxies). H2O uses accumulated received attention; FastV is a *context-only proxy* (canonical FastV is prompt-conditioned); LOOK-M degenerates to H2O on image-only pages; KVzip is a `recv_max` proxy for the reconstruction pass of Kim et al. (2025). On these cells the whole family sits *at or below random* page recall, so beating it is not a claim we make — they are query-blind controls, retained for completeness, and a faithful-reconstruction KVzip would likely be stronger. The honest comparators are `paq_stale`, `gold_frequency_static`, SnapKV-stale/fresh, and the retrieval arms.

On `gold_frequency_static`. This arm keeps the top-B pages by *gold-page frequency* across the document's queries. It uses gold labels, so it is not deployable; it is a *strong gold-informed static* *commitment* — but it is **not** provably accuracy-optimal and we do not call it "best possible" or a "ceiling" (an earlier draft did; the claim was retracted under review, Appendix B).

Reference rows. Full-context and oracle-evidence are reported for calibration only. SnapKV-fresh (re-run per query) is a reference, not an upper bound; our port averages heads and layers before pooling and is weaker than canonical SnapKV.

4 Methods

4.1 PAQ: coarse-to-fine page routing

PAQ is training-free page routing over a frozen VLM:

1. **Prefill once (per document).** Pages are rendered at coarse resolution (≈3.5K visual tokens total, i.e. ≈70 tokens/page on a 50-page document) and prefilled once; the KV cache is retained. Attention statistics are captured by OOM-safe per-layer forward hooks (Algorithm 4).

2. **Re-select per query.** The question is appended to the retained cache; the question rows' attention over page tokens (mean over heads, layers, and question tokens) is averaged over each page's token span to give a per-page score (Algorithm 1). Keep the top pages at budget b.

3. **Reread.** The selected pages are re-rendered at full resolution and the question is answered from them.

Scope, stated plainly: this is *page routing* — coarse-score → pick pages → full-resolution reread. It is *not* token-level KV compression: we never decode from a masked retained cache, and we claim no memory savings in v1. The estimand is query-conditioned page selection at matched page budget.

4.2 The reversibility framing and the isolation protocol

"Query-aware selection beats query-agnostic eviction" is both a known text result (Tang et al., 2024) and a strawman risk (FastV and LOOK-M can attend to the question). We therefore fix the axis to **reversibility**: every baseline runs its canonical scorer but is deployed *stale* — committed once per document (query-aware members commit an aggregate over the document's first $M = \min(3, k)$ queries) — while PAQ re-selects per query (Algorithm 2).

Reversibility is credited *only* if PAQ beats **both**

- `paq_stale`: PAQ's *own* scorer committed once. Identical scorer, identical budget; the only difference is reversibility, so the delta isolates the mechanism;

- `gold_frequency_static`: the strong gold-informed static, so the win is not an artifact of a weak committed selector.

This gate exists because a first, weaker gate (PAQ vs. best-of-eviction-family) *passed and was then retracted*: adversarial review showed the family's stale arms were weak-static artifacts (a gold-informed static nearly matched PAQ in recall), so "reversibility" had not been isolated. The two-comparator gate above was frozen before the confirmatory run (Appendix B documents both review rounds).

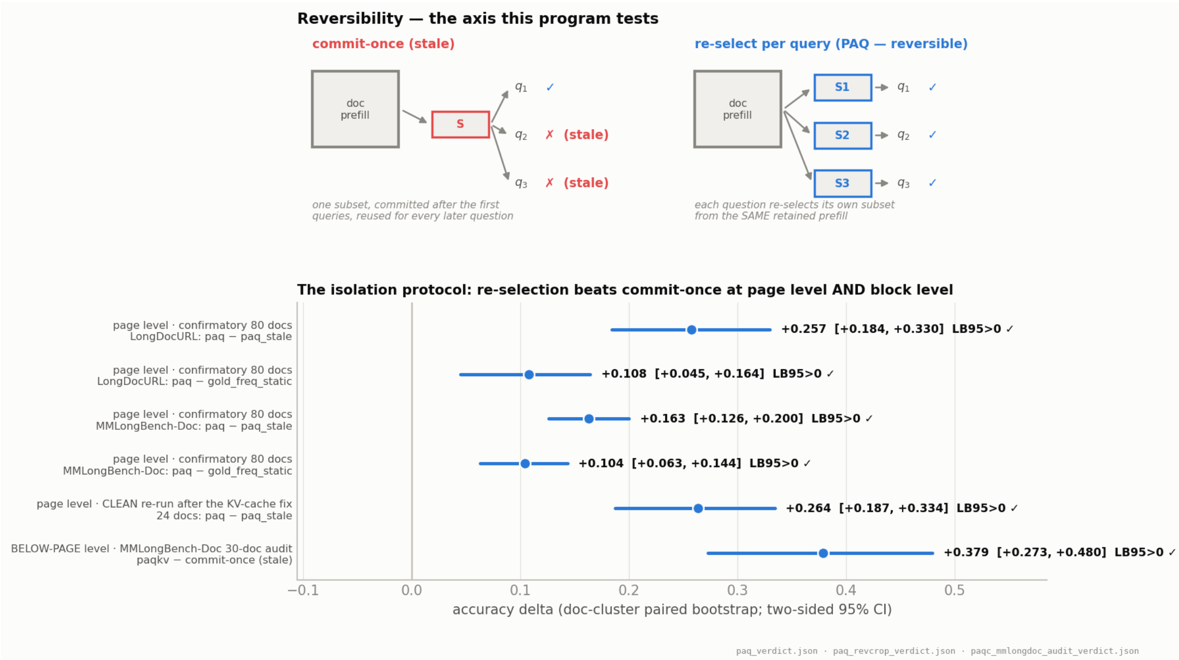

Figure 1: Reversibility — the tested axis. **Top:** a commit-once deployment reuses one subset for every question and goes stale; re-selection re-derives the subset per query from the same retained prefill. **Bottom:** the isolation protocol across granularities: the confirmatory page-level contrasts (PAQ − `paq_stale` and PAQ − `gold_frequency_static`, per cell, 80 documents/cell), the clean re-run after the KV-cache fix (§4.3), and the block-level PAQ-KV − commit-once contrast from the MMLongBench-Doc audit (§7.5.1). Every interval clears zero.

Figure 1 shows the contrast schematically together with every executed isolation interval — the page-level confirmatory contrasts, the clean post-KV-fix re-run, and the block-level PAQ-KV contrast: the reversibility benefit is not an artifact of one scorer, one granularity, or one bug regime.

4.3 Methods integrity: the KV-cache pollution bug and its fix

A cross-family build review of the PAQ-T read-eval harness (2026-07-16) found a real bug in the per-query scoring loop used across this program: `transformers` 5.x *mutates the retained* `DynamicCache` *in place*, and the loop reused the document prefill across a document's queries *without* restoring it (`past.crop(n_ctx)`) after each question forward. Query q_i therefore scored while attending to $q_0 \dots q_{i-1}$ in addition to the document. The bug was present in the PAQ-T recall gate *and* in the confirmatory scorer (`paq.py`, `HookedBackend.query_scores`) — i.e. in the run behind §7.1. The fix is the crop call after every per-query preview; it is now applied everywhere.

Its impact was quantified rather than assumed:

- **Blast-radius diagnostic** (`paq_kvcrop_diag.json`; 6 docs/cell, 87 query-rows): the bug is not inert — for a *fixed* scoring head it inflates gold-page recall by ≈+0.05 and shifts selections (mean top-B Jaccard clean vs. polluted 0.82), with the drift growing in the query index (Jaccard 1.00 → 0.61 from q_0 to q_6), as expected for cumulative cache contamination.

- **The recall gate was regenerated CLEAN** before any accuracy read: still `PAQ_T_VIABLE`; held-out expert recall 0.683/0.626 (vs. polluted 0.674/0.628), and the clean CV picks a more robust head (layer 21). All PAQ-T numbers in this document are from the clean gate.

- **The reversibility result was re-verified against the fix** (§7.1): on a 24-document/150-query clean-vs-polluted re-run, PAQ−`paq_stale` is +0.264 (LB_{95} = +0.187) clean vs. +0.284 polluted, and PAQ−`gold_frequency_static` +0.187 (LB_{95} = +0.137) clean vs. +0.228 polluted. The isolation contrast largely cancels the bug because both arms share the same (equally polluted) scorer; absolute PAQ accuracy in the confirmatory does carry some inflation (clean−polluted −0.041 on this check), which we disclose wherever absolute levels matter. The confirmatory reversibility *claim* stands.

Lesson recorded for the program: this is the second in-place-cache bug this codebase family has hit; every per-query preview over a retained cache now crops, and GPU smoke tests exercise the full path.

4.4 PAQ-T: expert-head scoring and the executed accuracy read-eval

After the retrieval loss (§7.2) we tested how much of the gap is an artifact of PAQ's mean-aggregation and resolution starvation (≈70 coarse tokens/page vs. ColQwen2's ≈1,024 patch embeddings/page). PAQ-T scores with (a) more scoring tokens per page (256 tokens/page, 6,000-token target total — a cap forced by encoder OOM on large documents, not full page resolution), and (b) a single *expert (layer, head)* chosen by held-out-document cross-validation (Algorithm 3) rather than the mean over all heads. Expert-head selection is CAPS's technique (Zhu et al., 2026); what is ours here is the amortized multi-query architecture it plugs into. "Training-free" means no parameter updates.

Claim scope (stated before the numbers). The head is chosen using gold page-recall on the complementary document fold *of the same benchmarks*. Cross-fitting removes row-level leakage (no query is scored by a head selected on its own document's fold), but this is **cross-validated, benchmark-supervised model selection — explicitly *not* zero-shot**. Removing this caveat is precisely the job of the B-lane adapter (§4.6).

The accuracy read-eval protocol (executed 2026-07-15/16; bars frozen before the run). The clean recall gate (§4.3) selects the expert heads: fold heads `21_11/21_18` on LongDocURL and `21_11/21_11` on MMLongBench-Doc — all in layer 21. Each held-out query is scored with the head trained on the *disjoint* fold (no selection-on-test); its top-B pages at $b = 5\%$ are then re-rendered at full resolution and read by the **same** sdpa reader, render policy, and scorer as the cached ColQwen2-fresh baseline rows — only the selected page set differs between arms. Inference is the document-cluster paired bootstrap of §5. Pre-registered bars (`paq_t_readeval_frozen_meta.json`): `NONINFERIOR` iff both cells' LB_{95}(PAQ-T−ColQwen2-fresh) > -0.03; `BEATS` iff pooled and both per-cell $\mathrm{LB}_{95} > 0$; `INFERIOR` iff either cell $\mathrm{LB}_{95} \leq -0.03$; per-cell power floors (150 queries, 15 documents) and an adversarial missingness imputation are part of the verdict. Because the high-resolution scoring pass OOMs on a handful of very large documents, coverage was completed by a *conservative* recovery pass: the missing documents were re-scored at a reduced 4,000-token scoring target — degrading only the PAQ-T arm's selection, never the cached baseline — bringing coverage to 463/463 and 440/444 (§7.4). Result: non-inferiority (parity), not a beat.

4.5 PAQ-KV: prefill-once, ask-many below-page KV selection (the central method)

PAQ-KV pushes the confirmed reversibility mechanism *below* page granularity — the operating point our literature search finds unoccupied (§8) — and is now the program's central method: it is

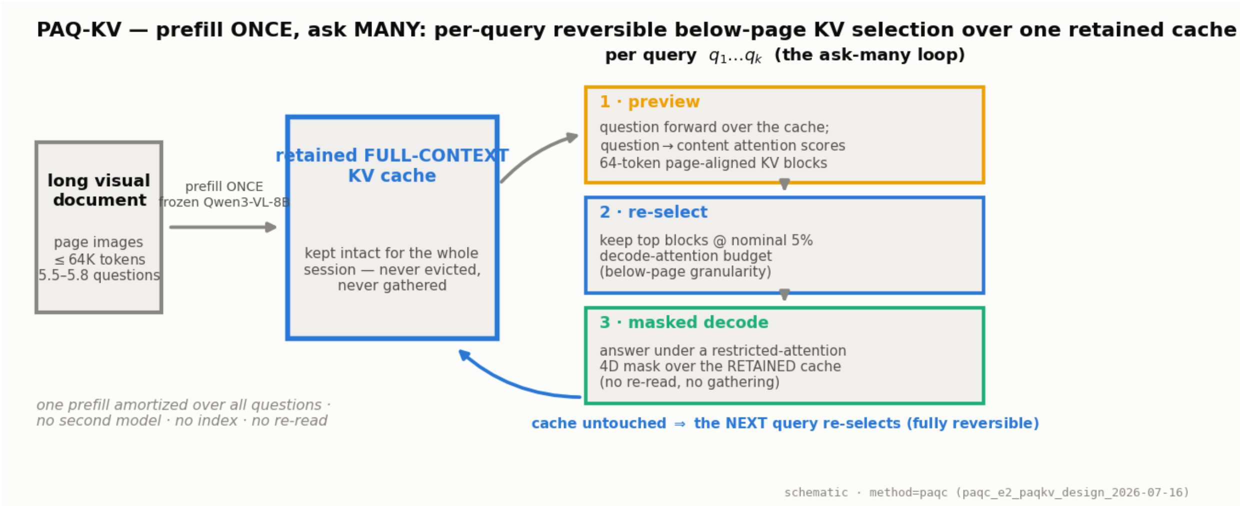

Figure 2: The PAQ-KV loop. The document is prefilled **once** into the frozen reader's KV cache, which is retained intact for the whole session. Each query then runs the same three steps — attention preview, block re-selection, masked decode — against that one cache. No second model, no index, no full-resolution re-read; the cache is never modified, so selection is reversible by construction.

what produces the MMLongBench-Doc beat of §7.5.1. Figure 2 shows the loop, Figure 3 the mask machinery, and Algorithm 5 the pseudocode.

Setting. One long visual document, read by a frozen Qwen3-VL-8B, will be asked k questions (5.5–5.8 gold-bearing per document in our pins). The document is prefilled **once** at the same rendering the cached full-context arm used ($\leq$64K tokens) and the KV cache is *retained*: never evicted, never physically compressed, never gathered. The cache plays the role a multi-vector index plays for a retriever — except it is the reader's own working memory, so "retrieval" and "reading" share one representation and one model.

Per query, three steps.

1. **Preview (score).** The tokenized question is forwarded over the retained cache — the crop-restored preview of §4.3, so queries never contaminate one another — and the question rows' attention over context positions (the same reader-internal signal used at page level) is aggregated into scores over **64-token, page-aligned KV blocks**: fine enough to isolate a table row, a figure caption, or a paragraph; coarse enough for the scores to be stable.

2. **Re-select.** Keep the top-scoring blocks at a **nominal 5% decode-attention budget**. Terminology deliberate: the kept blocks' K/V were computed *with full-document context* during the prefill (the standard claim semantics of the Quest/SnapKV method class), so this is a decode-attention budget, not a matched-token prefill comparison; the realized attended fraction on the executed screen is ≈5.9–6.0% of context, and is always reported. The matched quantity against ColQwen2 is the evidence budget (5% of pages vs. 5% of context) at identical reader and scorer.

3. **Masked decode.** The answer is generated while attending *only* to the selected blocks plus a small **always-keep set** (the BOS/system/template scaffold, the question itself, and previously generated tokens), enforced by a **4D additive attention mask** (batch, head, query row, key)

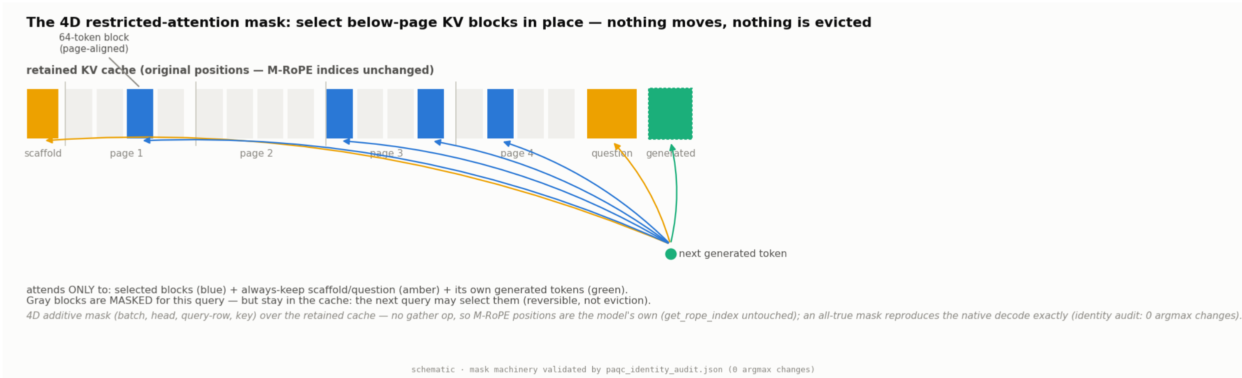

Figure 3: The mechanism. The retained cache is a strip of K/V at their original positions: scaffold, page content in 64-token page-aligned blocks, question, generated tokens. The 4D additive mask lets decode rows attend only to the per-query selected blocks (blue) plus the always-keep set (amber/green); everything else is masked *for this query only* — it stays in the cache and remains selectable by the next query. Because nothing is gathered, the M-RoPE position machinery is untouched, and an all-true mask provably reproduces the native decode (identity audit: zero mask-specific argmax changes).

over the retained cache. **No gathering**: the K/V tensors stay at their original positions, so every M-RoPE rotary index is exactly the model's own (`get_rope_index` unchanged) — the failure mode that makes physical cache-gathering treacherous for vision–language models is sidestepped entirely. No re-read, no re-render, no second model.

Exactness of the mask machinery. A decode-identity audit validated the mask path exactly: over 6 document–query pairs (32K–122K tokens), an all-true mask reproduces the native causal decode with **zero mask-specific argmax changes** (`paqc_identity_audit.json`); the earlier spike's single greedy-token divergence traced to `generate`'s mask-independent `is_causal` fast-path kernel, not to the mask. This audit was a pre-registered kill criterion for the lane.

Reversibility, now at block level. Because nothing is discarded, a selection is a *view*, not a commitment: the next query re-scores and re-masks the same complete cache. Commit-once eviction cannot do this — what it dropped is gone. The reversibility contrast of §4.2 therefore carries over verbatim, and it replicates at block level (PAQ-KV − commit-once = +0.379, $\mathrm{LB}_{95} = +0.273$; Figure 1).

What PAQ-KV is not. It is *not* CAPS (a separate scoring VLM, per-page full re-forwards, page granularity, single-query); *not* page-PAQ (no full-resolution re-read — the answer is decoded straight from the masked cache); and *not* eviction (the mask is per-query and reversible). The mask route is the *accuracy* path; physically gathering the cache (for bandwidth/memory savings) is a separate, future efficiency claim that we do not make.

Execution history (every gate frozen before its run). A 16-query two-cell spike passed; the decode-identity audit passed; a 30-document/cell development screen with commit-once, wrong-query, random-block, page-attend, and 2-D tile controls passed its ceiling-transfer, pooled-trajectory, and mechanism gates (its budget-curve monotonicity check *failed* on MMLongBench-Doc — accuracy at 10% is not above 5% there (0.398 vs. 0.402); disclosed, and consistent with selection quality rather than budget being the lever on that cell); both cells were then topped up to the full 80-document development pin (§7.5). A sharp-head scorer variant (C-V1) was separately gated and rejected (§7.6).

4.6 PAQ-T-LoRA / per-head mixture: the off-benchmark selection lane (B lane — executed, STOPPED)

The B lane attacked the parity result's two weaknesses at once — no beat, and benchmark-supervised head selection — by learning selection *off-benchmark*. The design (twice cross-family reviewed: REVISE/HOLD then HOLD, all blockers folded) was a toggleable LoRA on the reader's expert-layer Q/K projections, enabled only for the scoring forward; its cheapest pre-frozen gate came first: features were dumped for all $36 \times 32 = 1{,}152$ (layer, head) question→page attention channels, and a logistic **per-head mixture** was trained contrastively on off-benchmark, provenance-checked data (every training page perceptual-hashed against all 40,442 benchmark page images), with every design choice selected on external development data. The transfer gate then asked the only question that matters: does the externally trained mixture transfer to the benchmarks better than a *single* externally selected head?

It does not — STOP. On the development pin the mixture's gold-page recall is 0.652/0.594 (LongDocURL/MMLongBench-Doc) vs. 0.676/0.616 for the single external head: pooled mixture−single $= -0.023$ $[-0.037, -0.008]$ — significantly *worse* (`paqb_transfer_gate.json`, `GATE_PASS: false`, 907 queries). The 1,152-weight mixture overfits its off-benchmark training distribution; spreading weight over many heads loses to committing to one good head on transfer. Two things survive the stop: (i) the **byproduct** that the single best head selected with *no benchmark contact* is **L21.H18** — the same expert layer (and one of the same fold heads) the benchmark CV of §4.4 found, i.e. the expert head is a property of the model, not of the benchmarks; and (ii) the sealed-set discipline and provenance index, which the program keeps. The head byproduct was subsequently tested as a PAQ-KV block scorer (C-V1) and rejected there too (§7.6). No LoRA training was run beyond this gate; the lane is closed.

5 Definitions and statistical machinery

Matched page budget. For a document with n pages and budget fraction b, every arm keeps

$$B \;=\; \max\bigl(1, \lfloor b \cdot n \rfloor\bigr) \tag{1}$$

pages. Pages have unit budget weight in the executed vision harness, so this is the exact behavior of its cumulative-budget selector; ties in score are broken by page index. The same selector code path is used for every arm.

Gold-page recall at budget. For query q with nonempty annotated gold-page set G_q and kept set K_q ($|K_q| = B$),

$$\mathrm{recall@}b(q) \;=\; \frac{|K_q \cap G_q|}{|G_q|}. \tag{2}$$

Reported recall is the query-macro mean of this fraction within a cell; it is not answer accuracy and it is not the binary event "at least one gold page retrieved."

Commitment pressure. For a document with g distinct gold pages across its k queries,

$$\text{pressure} = \frac{g}{b \cdot n}, \tag{3}$$

i.e. how many times over the b-budget capacity the document's queries collectively need. This pre-registered quantity uses the fractional, pre-rounding capacity bn, whereas the executed selector uses B from Eq. (1). Thus pressure > 1 means $g > bn$; when $bn \geq 1$ this also implies $g > B$ and no single B-page subset can contain every distinct annotated gold page. For $bn < 1$ and at rounding boundaries, pressure should be read as a continuous diagnostic rather than a literal coverability test. This was the pre-registered mechanism axis; the formal test of it **failed** (§7.3) and it is retained as exploratory only.

Document-cluster paired bootstrap and LB_{95}**.** For arms A and C, let $d_q = s_A(q) - s_C(q)$ be their paired accuracy-score difference on query q. With two cells $c \in \{1, 2\}$ and N_c queries in cell c, the pooled point estimate is

$$\widehat{\Delta}_{A-C} = \frac{1}{2} \sum_{c=1}^{2} \left(\frac{1}{N_c} \sum_{q \in c} d_q \right). \tag{4}$$

Queries of one document share the document *and* the single stale subset, so bootstrapping over queries would be anti-conservative pseudoreplication. All intervals therefore resample *documents* with replacement within each cell (10,000 draws, seed 0), recompute Eq. (4), and take percentile endpoints. Per-cell intervals use the same procedure with one cell. We define

$$\mathrm{LB}_{95}(\Delta) = 2.5\text{th percentile of the bootstrap distribution of } \Delta, \qquad \mathrm{UB}_{95}(\Delta) = 97.5\text{th percentile}, \tag{5}$$

i.e. LB_{95} is the **lower end of a two-sided 95% CI** (equivalently a one-sided 97.5% bound). An earlier label of "one-sided 95%" was incorrect and has been fixed; the frozen gate arithmetic never changed.

The isolation gate (pre-registered, frozen before the run).

$$\textsc{Isolated} \iff \mathrm{LB}_{95}\big(\widehat{\Delta}_{\mathrm{PAQ-\texttt{paq\_stale}}}\big) > 0 \;\wedge\; \mathrm{LB}_{95}\big(\widehat{\Delta}_{\mathrm{PAQ-\texttt{gold\_frequency\_static}}}\big) > 0, \tag{6}$$

using complete cases at $b = 5\%$. The frozen gate is pooled with equal cell weighting; the per-cell intervals are corroborating analyses, not additional gate clauses. If either interval includes zero, the evidence is insufficient to isolate a reversibility benefit; that outcome would not establish equivalence between commit-once and per-query selection.

6 Algorithms

7 Results

The primary confirmatory numbers below come from `research/paq_verdict.json` (903 complete vision queries, 80 documents/cell, frozen v3 harness). The independent-extension, mechanism, and fusion diagnostics come from the documented post-hoc scripts in Appendix C; PAQ-T recall numbers come from the *clean, crop-fixed* `research/paq_t_gate_verdict.json` and PAQ-T accuracy numbers from `research/paq_t_readeval_verdict.json` and `research/paq_t_readeval_combined_verdict.js`

Algorithm 1 PAQ per-query scorer and matched-budget selection (vision cells)

Require: document pages $P_1, \ldots, P_n$; queries $q_1, \ldots, q_k$; budget b; frozen VLM (eager attention)

render all pages coarsely (≈3.5K visual tokens total); prefill once; retain KV cache C

record image-token spans $s_1, \ldots, s_n$ for the pages

for each query q_j **do**

  forward only the tokenized question against C (no re-prefill)

  via hooks: $\mathbf{a} \leftarrow$ mean over layers, heads, and question rows of attention to each context position ▷ Algorithm 4

  per-page score $\mathrm{score}_j(p) \leftarrow |s_p|^{-1} \sum_{t \in s_p} \mathbf{a}[t]$ ▷ `attn_max` variant: max over s_p

  $K_j \leftarrow$ top-B pages by score_j, $B = \max(1, \lfloor b\, n \rfloor)$ ▷ Eq. (1); index tie-break

  re-render pages K_j at full resolution; answer q_j from them

end for

Algorithm 2 Commit-once (stale) vs. re-select (fresh) deployments — the reversibility contrast

Require: per-query score vectors $v_1..v_k \in \mathbb{R}^n$ from ANY scorer (PAQ, SnapKV, ColQwen2)

fresh (reversible): for query q_j, select top-B from v_j ▷ one subset per query

stale (commit-once): $u \leftarrow \frac{1}{M} \sum_{j \le M} \mathrm{minmax}(v_j)$ with $M = \min(3, k)$; select top-B from u once; reuse that subset for ALL k queries

min-max normalization before averaging prevents any single query's magnitude from dominating the committed ranking; query order is frozen (arrival order).

query-agnostic arms (H2O/FastV/LOOK-M/KVzip/random/trunc) are stale by construction (their score vector never sees a question).

PAQ-KV numbers come from `research/paqc_topup_verdict.json` and `research/paqc_mmlo_topup_verdict.j` (80 documents/cell), `research/paqc_mmlongdoc_audit_verdict.json` (mechanism audit), and `research/paqc_scorer_verdict.json` (C-V1); the B-lane gate from `research/paqb_transfer_gate.json`. Every significance statement in this document is the doc-cluster paired bootstrap of §5 (`cluster_lb95` in `scripts/paq.py`) — the program's only valid instrument. Per-cell results are primary. "Pooled" inferential contrasts use the equal-cell statistic in Eq. (4); query-micro summaries are labelled separately.

7.1 Confirmatory isolation: reversibility is real

CONFIRMED Per-query re-selection beats the identical scorer committed once AND a strong gold-informed static, on both cells, document-clustered, surviving independent-extension and missingness worst-case checks.

Table 3 and Figure 4 give the gate result. At $b = 5\%$:

At screen scale (30 documents/cell), the MMLongBench-Doc contrast against the static was -0.017 with a CI including zero. It became positive at 80 documents; we treat that earlier result as an underpowered screen, not as an independent replication.

Independent extension (the screen is embedded in the confirmatory). Because the 80-document pin contains the 30 screen documents, we re-ran the isolation statistics on the **50 new documents per cell only** ($n = 582$ queries): PAQ − `paq_stale` $+0.216$ $[+0.159, +0.269]$; PAQ − `gold_frequency_static` $+0.108$ $[+0.059, +0.150]$; per-cell static contrasts $+0.089$ $[+0.004, +0.162]$

Algorithm 3 Expert-head selection with held-out-document CV (PAQ-T; gold-supervised selection on the complementary fold, never on the scored documents)

Require: per-(layer,head) question→page scores; gold sets G_q; documents D of one cell; budget $b = 5\%$

split D into two folds by document (seed-0 shuffle)
for each fold F **do**
    $Q_{\text{train}} \leftarrow$ queries in $D \setminus F$
    on Q_{train}: pick $(\ell^*, h^*) = \arg\max_{\ell,h} |Q_{\text{train}}|^{-1} \sum_{q \in Q_{\text{train}}} \text{recall@}b_{\ell,h}(q)$ ▷ Eq. (2)
    evaluate (ℓ^*, h^*) on the held-out fold F
end for
report query-macro held-out recall per cell ▷ no weight updates; head choice does use training-fold gold labels

Algorithm 4 OOM-safe per-layer-hook attention capture (the engineering that makes 64K scoring feasible)

register a forward hook on every language-decoder self-attention module (vision encoder excluded)
run the chunked prefill / question forward with `output_attentions=False` under eager attention
inside each hook: catch the layer's $(1, H, q, k)$ attention tensor, reduce it immediately (fp32-accumulated column-sum for H2O, running max for KVzip, layer-2 last row for FastV, question-row mean for PAQ), and **discard** it
peak memory = ONE layer's attention. The naive all-layers `output_attentions=True` path OOMs (≈77 GB at 128K text) or crashes vision cuDNN.
a capture guard hard-fails if no hook observed a 4D tensor (prevents silently scoring zeros)

(LongDocURL) and +0.127 [+0.083, +0.173] (MMLongBench-Doc). The result therefore also holds on the out-of-screen extension.

Answerable subset (secondary). On queries where oracle-evidence scores exactly 1 ($n = 407$, 137 docs): PAQ 0.583 vs. `paq_stale` 0.182 (+0.395 [+0.324, +0.460]) and vs. the static 0.382 (+0.204 [+0.135, +0.266]). Reported secondarily because it conditions on a model outcome.

Missingness. 4 of 907 vision queries (MMLongBench-Doc document `mi_phone`, queries 2–5) are missing due to a per-document timeout. Imputing them worst-case in either direction leaves the gate verdict unchanged.

Robustness to the KV-cache fix (integrity check, 2026-07-16). The confirmatory run predates the discovery of the cache-pollution bug (§4.3), so the isolation contrast was re-verified with the fix in place: on 24 documents/150 queries re-run clean vs. polluted at the confirmatory's coarse configuration (`research/paq_revcrop_verdict.json`), PAQ−`paq_stale` is **+0.264** [$\text{LB}_{95} = +0.187$, $\text{UB}_{95} = +0.334$] *clean* vs. +0.284 polluted, and PAQ−`gold_frequency_static` is +0.187 [+0.137, +0.246] clean vs. +0.228 polluted. Because PAQ and `paq_stale` share the same scorer, the pollution largely cancels in the contrast; the bug inflated the headline delta by only ≈0.02. Absolute PAQ accuracy is inflated by ≈0.04 on this check (clean−polluted −0.041,

Algorithm 5 PAQ-KV: per-query reversible below-page KV-block selection with a 4D restricted-attention mask (no re-read, no gathering)

Require: document pages rendered as in the full-context arm ($\leq$64K tokens); queries $q_1, \dots, q_k$; nominal budget $b = 5\%$; frozen VLM

```
prefill the document ONCE; retain the full KV cache C (context length n_ctx); record page-aligned 64-token block spans B_1, ..., B_m and the always-keep scaffold set A
for each query q_j do
    forward the tokenized question over C; capture question→context attention via the per-layer hooks (Algorithm 4); crop C back to n_ctx    ▷ the fix of §4.3
    score(B_i) ← aggregated question-row attention mass on block B_i's positions
    S_j ← top blocks by score at the nominal budget b · n_ctx    ▷ realized fraction reported (≈5.9–6.0%)
    build the 4D additive mask M_j: decode rows may attend to A ∪ S_j ∪ q_j ∪ generated tokens; all other context positions get −∞
    decode the answer from C under M_j — no gathering: K/V positions and M-RoPE indices unchanged
    discard M_j; C is intact for q_{j+1}    ▷ reversibility: a selection is a view, not a commitment
end for
```

Table 3: Confirmatory isolation at b=5% on 903 complete queries (accuracy; document-cluster paired bootstrap). Contrast rows report $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$; the pooled column uses equal cell weighting.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
PAQ accuracy	0.385	0.258	0.322
`paq_stale` accuracy	0.128	0.095	0.111
`gold_frequency_static` accuracy	0.278	0.154	0.216
PAQ − `paq_stale`	+0.257 [+0.184, +0.330]	+0.163 [+0.126, +0.200]	+0.210 [+0.168, +0.250]
PAQ − `gold_freq_static`	+0.108 [+0.045, +0.164]	+0.104 [+0.063, +0.144]	+0.106 [+0.067, +0.141]
Verdict	REVERSIBILITY ISOLATED (both $\mathrm{LB}_{95} > 0$, both cells and pooled)		

[−0.077, −0.003]), which should be kept in mind when reading the absolute (non-contrast) rows of Table 3. **The confirmatory reversibility claim stands under the fix.**

Budget curves. Figure 5 shows accuracy at $b \in \{2.5, 5, 10\}$%: PAQ dominates `paq_stale` and the static at every budget on both cells, monotonically. On LongDocURL, PAQ at $b = 10\%$ reaches 0.518 vs. full-context 0.546 — reading 10% of pages recovers ≈95% of full-context accuracy on that cell (not on MMLongBench-Doc: 0.317 vs. 0.397).

7.2 The deployable retrieval comparison: PAQ loses at page level

> **CONFIRMED NEGATIVE** A strong trained visual retriever re-ranked per query beats PAQ by ≈10 accuracy points at matched page budget. This is the decisive result of the final review round, reported with the same prominence as the positive.

The adversarial AAAI-readiness review named one decisive missing experiment: a deployable query-conditioned *retrieval* baseline. We ran it: ColQwen2 (Faysse et al., 2025) (late-interaction MaxSim over page images; pages embedded once per document) fresh and stale, plus CLIP (Radford

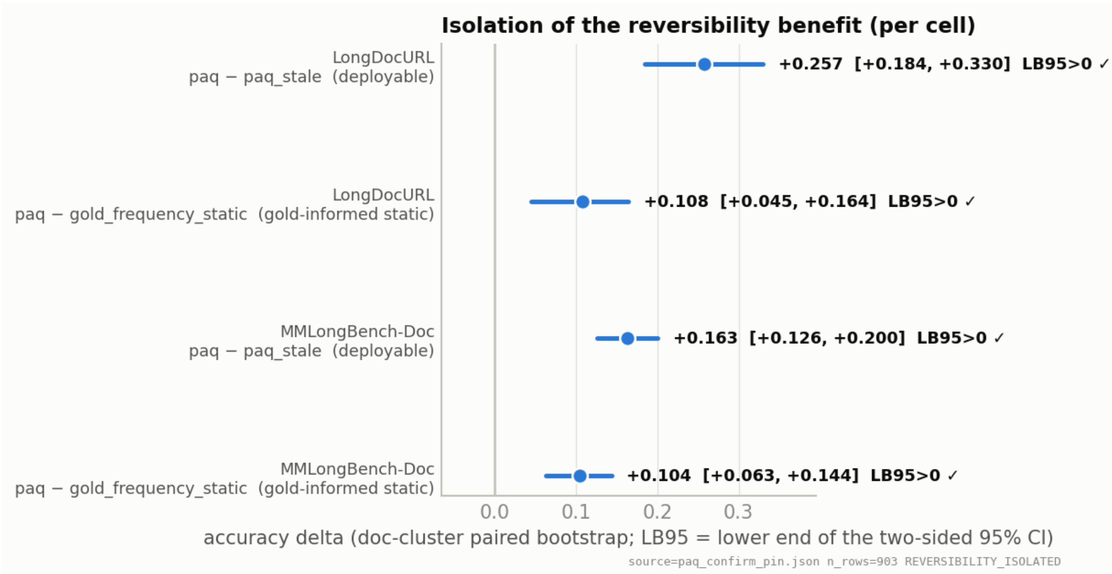

Figure 4: The statistical headline: per-cell forest of the two isolation contrasts at $b = 5\%$. Intervals are document-clustered two-sided 95% CIs; all four exclude zero.

Table 4: Deployable retrieval comparison at b=5% (accuracy; 903 complete queries). PAQ loses to ColQwen2-fresh on both cells. Fresh exceeds stale for both tested scorers. The pooled arm means use equal cell weighting; contrast intervals are document-clustered.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
ColQwen2-fresh	0.512	0.328	0.420
PAQ	0.385	0.258	0.322
ColQwen2-stale (commit-once)	0.216	0.138	0.177
CLIP-fresh	0.155	0.164	0.159
full-context (reference)	0.546	0.397	—
PAQ − ColQwen2-fresh	pooled -0.099 $[-0.129, -0.072]$ $\Rightarrow$ **PAQ loses**		
PAQ − ColQwen2-stale	pooled $+0.145$ $[+0.101, +0.185]$		
PAQ − CLIP-fresh	pooled $+0.162$ $[+0.128, +0.197]$		

et al., 2021) fresh, same reader, same budgets, all 903 queries. At $b = 5\%$, pooled, document-clustered (Table 4, Figure 6):

Three readings, all of which we stand behind:

1. **PAQ is not a superior page router.** The loss is ≈10 points, LB_{95} well below zero, and it holds on every stratum we examined (single-gold −0.110, multi-gold −0.086; no question type where attention beats retrieval).

2. **The reversibility pattern spans both tested scorer families.** On the query-micro summary, ColQwen2-fresh is 0.423 vs. 0.178 stale; the corresponding equal-cell means are 0.420 and 0.177. This reproduces the fresh-vs-stale collapse with a different scorer. It supports per-query reselection for attention- and retrieval-based systems in this evaluated regime; it is not a claim about every possible scorer.

3. **The two existing outputs have limited complementarity.** A per-query oracle taking the better observed answer score from {PAQ, ColQwen2-fresh} gains +0.035 pooled

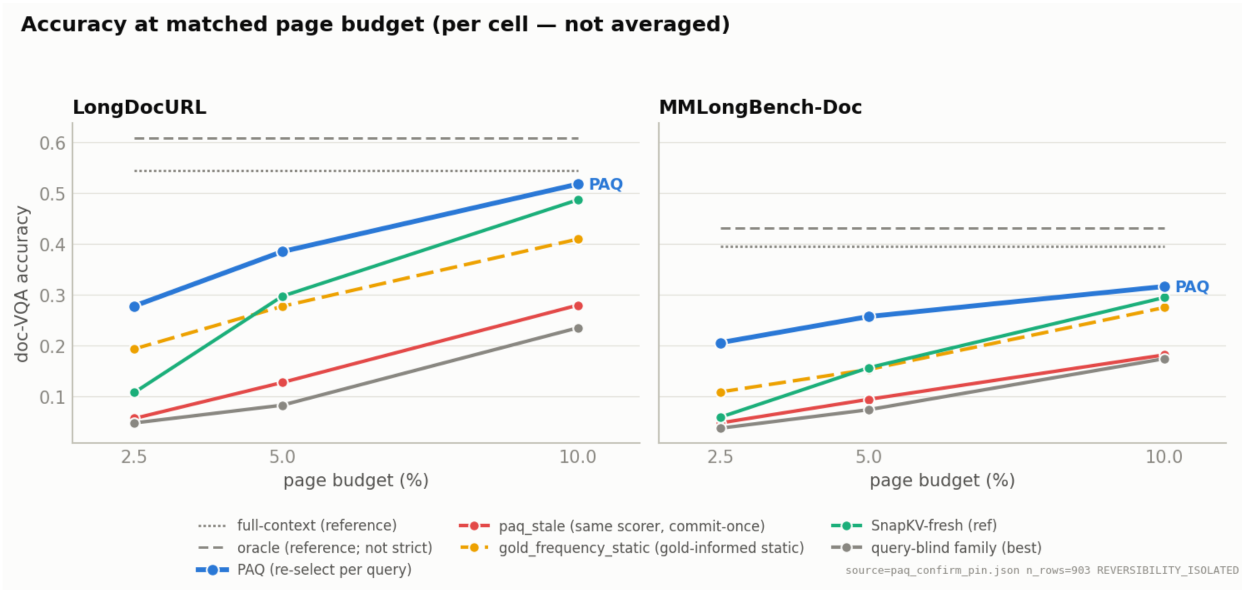

Figure 5: Accuracy at matched page budget, per cell (never averaged). Full-context and oracle-evidence are recessive reference lines, not competitors; the "query-blind family (best)" line is the demoted control family's best member.

over ColQwen2-fresh (LB_{95} +0.024; +0.041/+0.030 per cell). Among the 419 queries where ColQwen2-fresh scores exactly zero, PAQ scores one on only ≈6%. This makes simple output routing low priority. It does *not* upper-bound a fusion or reranker that selects a new page set, whose reader output could differ from both inputs.

Gold-page recall is consistent with a coverage contribution to the loss: at $b = 5\%$, ColQwen2-fresh recalls 0.664/0.536 (per cell) vs. PAQ 0.494/0.443. This observation motivated the PAQ-T selection diagnostic and read-eval (§7.4); by itself it is not a causal decomposition of the accuracy gap.

Why the loss redirects rather than ends the project. Simply replacing mean aggregation with a better page-level attention head is not a clean novelty route: CAPS has already published expert-head attention selection on these benchmarks (§8.1). The PAQ-T read-eval (§7.4) has since shown that the expert head recovers the whole ≈10-point gap — to *parity*, not past it — which is what redirected the program to the two lanes of §9.3.

7.3 Mechanism: demoted to exploratory

RETRACTED AS A CLAIM The pre-registered "commitment-pressure gradient" mechanism FAILED its formal test; what replaced it is a corrected, two-part account of where the benefit comes from.

We had hypothesized the reversibility benefit concentrates in high-commitment-pressure documents (Eq. (3)) and reverses under low pressure. The screen suggested this; the formal document-clustered interaction test on the confirmatory does **not** support it: high−low pressure difference +0.042 [−0.048, +0.130] on PAQ−static and −0.015 [−0.116, +0.080] on PAQ−`paq_stale` — both include zero. LongDocURL shows the predicted direction without significance; MMLongBench-

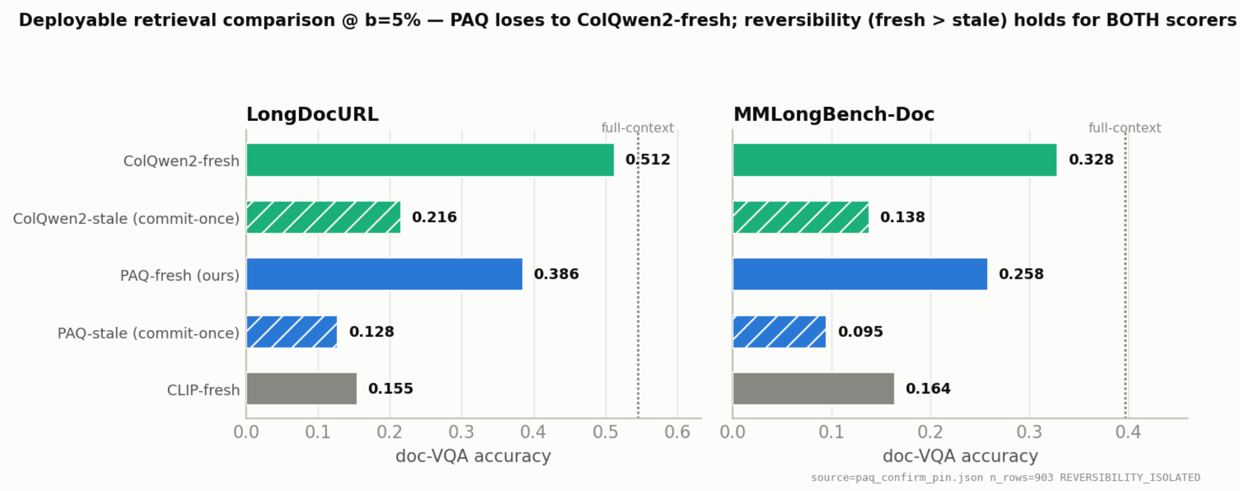

Figure 6: The two honest facts in one picture, per cell: (i) ColQwen2-fresh beats PAQ (the ≈10pt loss); (ii) *both* scorers collapse when committed once — reversibility is a general property of the multi-query regime, not a PAQ-specific effect.

Doc's benefit is broad across pressure. Figure 7 is therefore labelled exploratory and this document makes no mechanism-gradient claim.

An earlier "reading quality, not coverage" explanation was also **wrong**: PAQ's gold-recall exceeds the static's by +0.092 (the earlier "recall ≈ static" countercheck was in error), and at *matched* recall ($n = 415$) PAQ still beats the static by +0.049 $[+0.021, +0.076]$. The corrected descriptive statement is: **per-query re-selection has higher gold coverage overall and higher downstream accuracy among queries with equal measured gold recall**. The matched-recall analysis does not identify why the selected page sets differ or establish a separate causal "reading quality" mechanism.

7.4 PAQ-T: accuracy read-eval — non-inferiority (parity), not a beat

EXECUTED — PARITY The pre-registered accuracy read-eval verdict is `PAQ_T_ACC_NONINFERIOR`: PAQ-T *matches* ColQwen2-fresh at matched page budget, training-free, with no separate model — but does **not** beat it. The head selection is benchmark-supervised (CV), and the expert-head technique is CAPS's.

Recall gate (clean, crop-fixed). The PAQ-T gate (score-only; no reads) asked whether the recall deficit behind the −10pt loss is an artifact of mean aggregation and scoring resolution. Regenerated clean after the KV fix (§4.3), its verdict is `PAQ_T_VIABLE`: held-out-document expert-head recall@b=5% is **0.683** (LongDocURL, $n = 451$) and **0.626** (MMLongBench-Doc, $n = 425$) vs. ColQwen2's in-harness 0.664 and 0.536 — PAQ-T recall meets or exceeds ColQwen2 on both cells (+1.9/+9.0pt). The expert head is the lever (mean aggregation: coarse 0.494/0.443, higher-resolution 0.500/0.452); resolution contributes almost nothing. Selected fold heads are `21_11/21_18` (LongDocURL) and `21_11/21_11` (MMLongBench-Doc), all layer 21 (Table 5).

The accuracy read-eval (pre-registered, frozen bars; protocol in §4.4). Table 6 gives the result at near-full coverage (463/463 LongDocURL with 12 OOM'd queries recovered; 440/444 MMLongBench-Doc with 19 recovered; the 4 `mi_phone` queries remain unpairable):

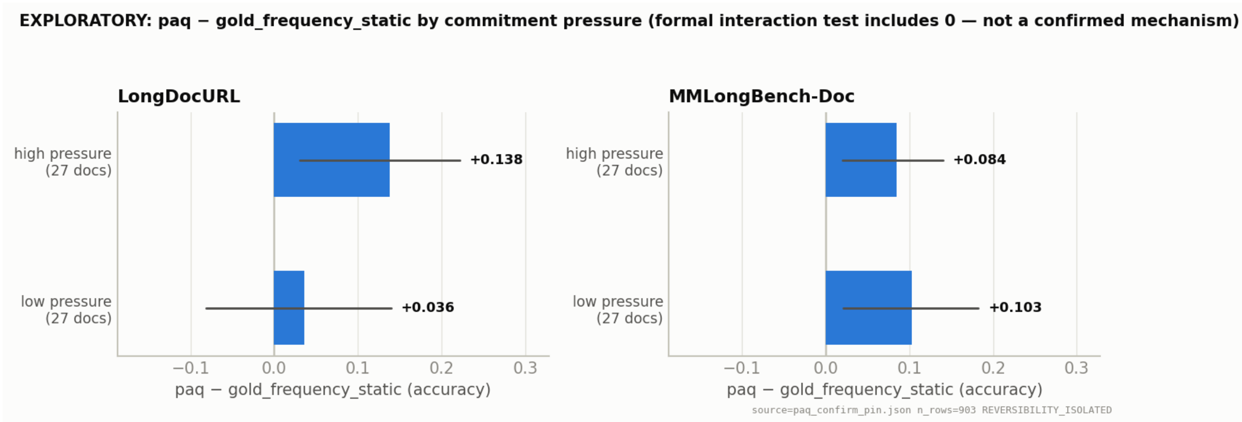

Figure 7: EXPLORATORY (demoted): PAQ−`gold_frequency_static` by commitment-pressure split. The formal high−low interaction includes zero on both contrasts; shown for completeness, not as a confirmed mechanism.

Table 5: PAQ-T page-selection recall@b=5% diagnostic (*clean*, crop-fixed gate). Sample sizes differ slightly across rows, so they should not be read as paired deltas.

Scorer	LongDocURL	MMLongBench-Doc
coarse mean aggregation (PAQ v1)	0.494 ($n = 463$)	0.443 ($n = 440$)
higher-resolution mean aggregation	0.500 ($n = 451$)	0.452 ($n = 425$)
expert head (held-out-document CV)	**0.683** ($n = 451$)	**0.626** ($n = 425$)
ColQwen2 in-harness reference	0.664 ($n = 463$)	0.536 ($n = 440$)

The missingness story matters and we tell it in full. The *complete-case* verdict (872 paired queries) looked slightly better: pooled +0.023 [$\mathrm{LB}_{95} = +0.005$], mmlongdoc +0.039 [+0.014]. The result cross-review (Sol) blocked it for **informative missingness**: the ≈4% of queries whose high-res scoring OOM'd (6 big documents) have cached ColQwen2 accuracy 0.62 — far above the complete-case cell means (0.51/0.32) — so dropping them optimistically biased the comparison toward PAQ-T, and the adversarial imputation failed the non-inferiority bar. Recovering 31 of the 35 missing rows (conservative lower-res pass, §4.4) resolved the block and **erased the pooled edge**: the honest, near-full-coverage headline is the parity row above, with pooled $\mathrm{LB}_{95} = -0.000$. The reviewer was right; the complete-case numbers are retained only as this paragraph.

Secondary contrasts (near-full coverage, 872-query paired set). PAQ-T beats the original coarse PAQ by +0.119 [+0.090, +0.150] and high-resolution mean aggregation by +0.101 [+0.073, +0.132] — the expert head, not resolution, is the accuracy lever. PAQ-T sits only −0.035 [−0.065, −0.007] below full-context. ColQwen2-fresh > coarse-PAQ is reconfirmed at +0.097 [+0.069, +0.127].

What this is and is not (blunt). A training-free, single-model, reader-internal selector now *matches* a SOTA trained visual retriever on end-to-end accuracy at matched page budget, in the amortized multi-query regime — with no separate retriever model and no multi-vector index. That is a real property, and it discharges the "within 2–3 points" non-inferiority bar predeclared by the efficiency red-team. It is **not** the beat this program is mandated to deliver: the pooled LB_{95} sits exactly at zero, LongDocURL is parity, and the mmlongdoc per-cell $\mathrm{LB}_{95} = +0.001$ is a razor's edge

Table 6: PAQ-T accuracy read-eval at b=5%, near-full coverage (903 paired queries; document-cluster paired bootstrap; contrast rows $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$ where available). Verdict: `PAQ_T_ACC_NONINFERIOR` — both cells clear the pre-registered $\mathrm{LB}_{95} > -0.03$ margin; the `BEATS` bar (all $\mathrm{LB}_{95} > 0$) is *not* met.

	LongDocURL	MMLongBench-Doc	pooled (equal-cell)
PAQ-T accuracy	0.518	0.357	—
ColQwen2-fresh accuracy	0.512	0.328	—
PAQ-T − ColQwen2-fresh	$+0.006\ [-0.018, +0.030]$	$+0.028\ [+0.001, +0.055]$	$+0.017\ [-0.000, +0.035]$

Table 7: PAQ-KV at the 80-document development pin, nominal 5% budget (doc-cluster paired bootstrap; contrast rows $\Delta\,[\mathrm{LB}_{95}, \mathrm{UB}_{95}]$). **All accuracy rows are the verified 80-document dev-pin means** — the LongDocURL full-context value is the 80-document dev value 0.5456 (an earlier draft used the 30-document screen value 0.514 here, which wrongly implied PAQ-KV beat full-context on LongDocURL; corrected). On LongDocURL PAQ-KV is *below* full-context (−0.014, not significant). The MMLongBench-Doc oracle/full contrasts pair on 440 of 444 queries (the 4 `mi_phone` rows have no cached reference, §7.1).

	LongDocURL	MMLongBench-Doc
PAQ-KV@5% (ours)	**0.5315**	**0.4042**
ColQwen2-fresh (matched budget)	0.5124	0.3254
full-context (reference)	0.5456	0.3965
gold-page oracle (reference; not strict)	0.6092	0.4314
PAQ-KV − ColQwen2-fresh	$+0.019\ [-0.019, +0.059]$	$+0.079\ [+0.041, +0.116]$
PAQ-KV − full-context	−0.014 (n.s.)	$+0.009\ [-0.022, +0.040]$
PAQ-KV − gold-page oracle	—	$-0.026\ [-0.059, +0.008]$
Queries (decoded / paired)	463 / 463	444 / 440
Verdict (pre-frozen bar: per-cell $\mathrm{LB}_{95} > 0$)	HOLD	**beat holds**

we refuse to sell as a win. Two caveats carry over: the head selection is benchmark-supervised CV (scope in §4.4), and the selector technique is CAPS's — our residual novelty remains the amortized one-prefill-many-queries architecture.

7.5 PAQ-KV at power: a beat on MMLongBench-Doc, a characterized hold on LongDocURL

CONFIRMED (DEV PIN) / HOLD At the full 80-document development pin, PAQ-KV@5% **beats** ColQwen2-fresh on MMLongBench-Doc (+0.079, $\mathrm{LB}_{95} = +0.041$ — the pre-frozen per-cell bar clears) and *matches* the gold-page oracle at that budget; on LongDocURL it matches ColQwen2 (+0.019, $\mathrm{LB}_{95} = -0.019$) and does not clear. The sealed one-touch final has not run.

Table 7 and Figure 8 summarize. Both cells use the frozen doc-cluster paired bootstrap (the program's only significance instrument), the cached ColQwen2-fresh/full-context/oracle rows of the confirmatory panel, and a PAQ-KV arm byte-identical to the frozen screen configuration (config-hash checked); the top-ups added the 50 previously unscored pin documents per cell without touching the sealed set. The pre-frozen bar was per-cell $\mathrm{LB}_{95}(\text{PAQ-KV@5\%} - \text{ColQwen2-fresh}) > 0$.

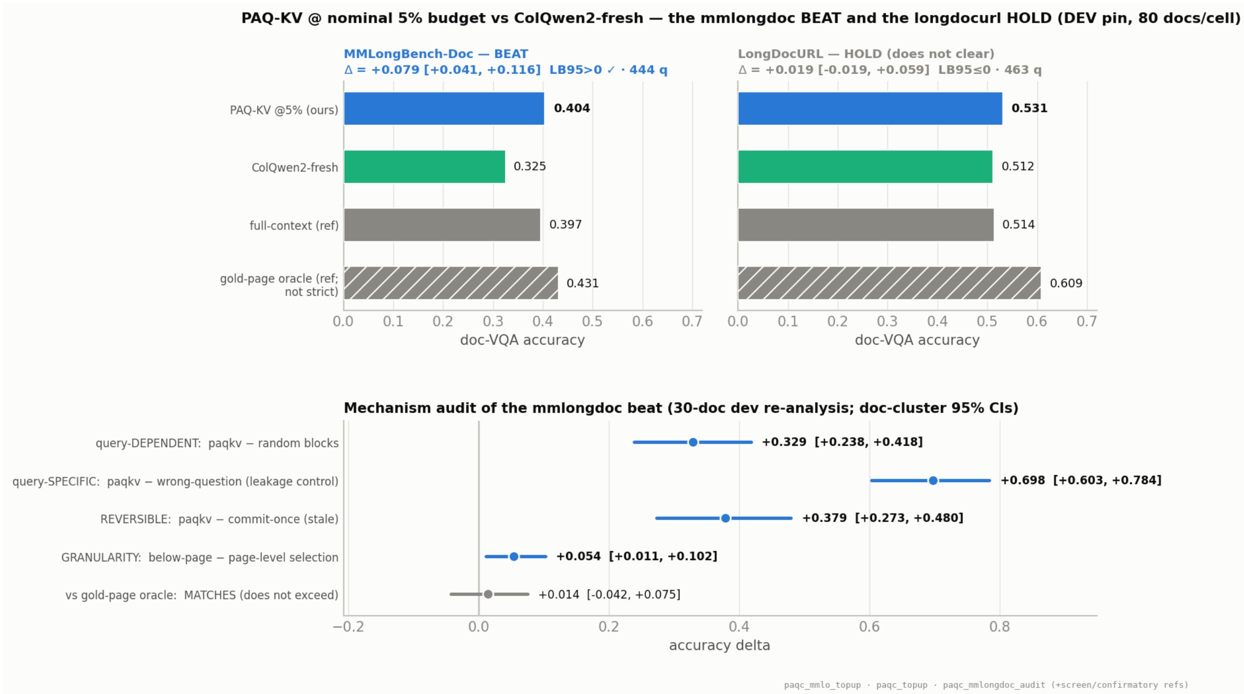

Figure 8: The current results panel. **Top:** PAQ-KV@5% vs. ColQwen2-fresh with the full-context and gold-page-oracle references, per cell, with the doc-cluster CIs: the MMLongBench-Doc beat clears ($\text{LB}_{95} = +0.041$); LongDocURL holds ($\text{LB}_{95} = -0.019$). **Bottom:** the mechanism audit of the beat (30-document dev re-analysis): selection is query-dependent, query-specific (the wrong-question collapse rules out leakage), reversible, and below-page > page-level; against the gold-page oracle PAQ-KV *matches* — the CI straddles zero, and we do not claim to exceed it.

7.5.1 MMLongBench-Doc: the beat, and its audit

The result. At 80 documents / 444 queries, PAQ-KV@5% scores 0.404 vs. ColQwen2-fresh 0.325: $+0.079$, $\text{LB}_{95} = +0.041$ — the pre-frozen per-cell bar clears with margin (`paqc_mmlo_topup_verdict.json`, `CALL: BEAT_HOLDS`). Against the references, PAQ-KV attending to a twentieth of the context *matches* full-context ($+0.009$ [$-0.022, +0.040$]) and *matches* the gold-page oracle (-0.026 [$-0.059, +0.008$]).

Correction of an earlier over-claim. An earlier draft of this program's notes said PAQ-KV "beats the gold-page oracle" on this cell. The adversarial audit (`paqc_mmlongdoc_audit_verdict.json`) showed that to be a 30-document artifact: the contrast straddles zero at both scales (30-doc $+0.014$ [$-0.042, +0.075$], per-query wins/ties/losses 14/126/17; 80-doc -0.026 [$-0.059, +0.008$]). The correct statement — still the striking one — is that PAQ-KV **matches the gold-page oracle at a 5% budget**: parity at a fraction of the budget, not a beat of the oracle. The oracle is defined identically on both cells (read exactly the annotated gold pages) and was pre-registered as *not* a strict ceiling.

The mechanism audit (30-document dev data, doc-cluster CIs; Figure 8, bottom). The beat survives every mechanistic adversarial check available on existing data:

- **query-dependent**: PAQ-KV − random-block selection = $+0.329$ [$+0.238, +0.418$];
- **query-specific**: PAQ-KV − wrong-question selection (select with a *different*, disjoint-gold

query's blocks; decode the true question) = +0.698 [+0.603, +0.784] — the collapse rules out leakage and query-invariant shortcuts;

- **reversible**: PAQ-KV − commit-once (stale) = +0.379 [+0.273, +0.480];
- **granularity**: below-page − page-level selection at matched budget = +0.054 [+0.011, +0.102] — the granularity itself, not just query-awareness, carries accuracy on this cell.

Caveats, kept visible. The audit contrasts are a re-analysis of 30-document development data (the beat itself is at 80 documents; the audit's named residual risk — sample size — was retired by the top-up). The screen's budget-curve monotonicity gate failed on this cell (10% is not above 5%: 0.398 vs. 0.402 — selection quality, not budget, moves this cell). And everything is development-pin evidence pending the sealed one-touch (§9.3).

7.5.2 LongDocURL: the honest limitation, characterized

The result. At 80 documents / 463 queries, PAQ-KV@5% scores 0.531 vs. ColQwen2-fresh 0.512: +0.019, $LB_{95} = -0.019$ — a positive point estimate that does **not** clear the pre-frozen bar (`paqc_topup_verdict.json`, `CALL: HOLD`). We report this as a genuine, well-characterized limitation, not a failure of the mechanism: the mechanism checks (query dependence, reversibility) clear on this cell too; what is missing is the *margin*.

Why the cell resists selection methods. A CPU re-analysis of the existing rows (design note `paqc_longdocurl_extensions_2026-07-17.md`, cross-family reviewed with corrections folded in below) characterizes the shortfall:

1. **The significance bar is ≈twice the observed delta.** At 80 documents the per-document delta s.d. is 0.173, giving a planning-level bootstrap s.e. of ≈ 0.020: an observed $LB_{95} > 0$ needs $\Delta \gtrsim +0.039$. (The review [Sol] correctly notes this symmetric-normal figure is only a planning approximation for a percentile bootstrap and is method-dependent; we use it as a compass, not a gate.)
2. **Only ≈10% of queries are selection-capturable.** Joint failure taxonomy over the 463 rows: 48.4% are answered under both PAQ-KV and the gold-page oracle; **39.1% fail under both** — these queries score zero even when the reader is handed exactly the gold pages at native resolution, so no selection method of any kind can rescue them (a *reader-conversion wall*, ≈41% of the cell counting the oracle-fail mass); 10.4% are oracle-right/PAQ-KV-wrong (the entire selection-capturable pool, worth +0.05 net if captured perfectly); 2.2% have PAQ-KV beating the oracle.
3. **Resolution is not the lever.** A CPU audit of the pin's images shows native page resolution is *below* the prefill per-page token cap for 79/80 documents (median native/cap = 0.50): the prefill and the oracle both already see native resolution, and the corpus ships JPGs with no higher-DPI source. The oracle−PAQ-KV gap is selection/conversion, not resolution.
4. **Large documents are a low-leverage dead zone.** Per-document mean(PAQ-KV−ColQwen2): +0.064 on the smaller half (25–44 pages) vs. −0.005 on the larger half (44–102 pages), where every arm is flat (≈0.48 at any budget) and even the native-resolution gold-page oracle reaches only 0.534. The method's edge lives on small/medium documents; the large-document mass adds variance without signal.

5. **Pure-selection ceilings are at best knife-edge.** Measured per-query oracle-chooser unions cap at or below the bar: best-of-budgets +0.023; tile$\vee$stripe geometry +0.024; the two-scorer union V0$\vee$V1, run through the frozen instrument under review, does clear — but only at $\text{LB}_{95} = +0.003$, and it requires an *undeployable* per-query oracle chooser. The only measured lever with comfortable headroom is a PAQ-KV $\cup$ ColQwen2 combination (union oracle $\text{LB}_{95} = +0.061$ vs. ColQwen2), which we exclude here: a retriever-in-the-loop ensemble is not the standalone method this program is mandated to ship, and the cross-family review classified it as exactly that ("a hybrid system beats standalone ColQwen2" is a different claim).

In short: on LongDocURL the binding constraint is the reader's evidence-to-answer conversion and the cell's variance structure, not selection quality. That is a finding about *where* below-page reversible selection helps (dispersed-evidence cells like MMLongBench-Doc) and where it cannot (conversion-bound cells), and we frame it as such.

7.6 Lane negatives: the B-lane STOP and the C-V1 no-go

NEGATIVES Both attempted selector upgrades failed their pre-frozen gates and are reported as ablations: an externally learned per-head mixture does not transfer (B), and a sharper recall-optimized head does not convert to accuracy (C-V1).

B lane (off-benchmark per-head mixture): STOP. Full account in §4.6: the externally trained 1,152-head mixture transfers *worse* than a single externally selected head (pooled recall delta -0.023 $[-0.037, -0.008]$; `GATE_PASS: false`). The lane's byproduct is real: head **L21.H18**, selected with no benchmark contact, coincides with the benchmark-CV expert layer of §7.4 — evidence that the expert head is a property of the model, not of the benchmarks (this removes the "benchmark-supervised" caveat from the head's *existence*, not from any benchmark-tuned use of it).

C-V1 (the sharp head as PAQ-KV's block scorer): NO-GO. Swapping the diffuse mean-attention block scorer (V0) for the transfer-validated sharp head L21.H18 (V1) was gated on three pre-frozen conditions; none held (`paqc_scorer_verdict.json`). On LongDocURL, V1 nudges the point estimate ($0.531 \to 0.536$; V1$-$ColQwen2 $= +0.024$ $[-0.012, +0.066]$) but does not separate from V0 ($\text{LB}_{95}(\text{V1} - \text{V0}) = -0.011$, n.s.); on MMLongBench-Doc it *hurts* ($0.402 \to 0.381$ on the screen documents). The lesson generalizes the program's recurring theme: **recall-optimized selection is not accuracy-optimized selection** — a sharper, retrieval-style scorer concentrates budget on what looks relevant to a ranking objective and loses the contextual slack the diffuse scorer keeps. The LongDocURL point estimate ($\approx +0.02$ over ColQwen2) is real but sub-significant, and no measured selection variant converts it into a clearing margin (item 5 of §7.5.2).

7.7 Efficiency framing (measurement pending)

UNMEASURED No latency/VRAM/throughput table exists yet. Everything in this subsection is an accounting argument, flagged as such, with the red-team's caveats folded in.

PAQ is training-free, uses no separate retriever model and no multi-vector index; the coarse prefill is paid once per document and each query costs one short forward over the retained cache plus the reread. But the honest accounting (from the efficiency red-team) is: ColQwen2 *also* amortizes (page embeddings are computed once; a query costs a small query encoding plus MaxSim), and PAQ's per-query readout is a reader-scale forward — likely *more* per-query FLOPs than ColQwen2's query encoding. What PAQ genuinely saves is *dependencies*: no second model resident, no index storage, one deployed artifact. Where the amortization argument has real teeth is against CAPS, which re-forwards every page per query (11.25 s/query reported) and explicitly concedes the multi-query regime to embedding methods. An efficiency-only justification while losing ≈10 accuracy points was reviewed as weak-reject; we therefore make no efficiency claim until (a) the measured table exists (cold/warm start, retriever co-resident vs. sequential, index bytes/page, several queries-per-document), and (b) PAQ-T accuracy lands within the predeclared non-inferiority bars (within 2–3 points, $\mathrm{LB}_{95} > -0.03$, both cells). Condition (b) is now met (§7.4: `PAQ_T_ACC_NONINFERIOR`); condition (a) — the measured table — still does not exist, so this subsection remains an accounting argument and no efficiency claim is made.

PAQ-KV changes the shape of the accounting. PAQ-KV never re-reads and never re-renders: after the one prefill, a query costs one short question forward (the preview) plus a masked decode over an already-resident cache, while ColQwen2's amortized regime still pays a per-query prefill of the re-read selected pages. That is a structurally stronger amortization story than page-PAQ's — and on MMLongBench-Doc it now comes attached to an accuracy *beat* rather than a deficit (§7.5.1). It remains an accounting argument: the measured table (latency, VRAM, index bytes, queries-per-document break-even) still does not exist, and no efficiency claim is made.

8 Related work

8.1 CAPS: the page-level method is already published

CAPS (Zhu et al., 2026) ("Attention as Selector", ACL 2026) publishes attention-based page selection for long-document retrieval *on our exact benchmarks*: a small scoring VLM's final-token→page attention at a single expert (layer, head), lightly calibrated on 6.4k off-benchmark contrastive triplets, beats ColPali-pipeline retrieval (MMLongBench R@5 78.89 vs. 72.00; downstream QA 44.69 vs. 36.18–41.01). Their training-free ablation lands just *below* ColPali (best zero-shot head R@5 70.92 vs. 72.00) — independently corroborating our own training-free loss. We state plainly: **the "reader-attention selects pages" method is scooped**, and any contribution of ours must be argued elsewhere. CAPS's own concessions define the open ground: it scores every page independently per query (multi-query latency conceded to embedding methods — exactly PAQ's amortized regime) and operates at page granularity only (region-level selection named as an open problem — exactly extension E1).

8.2 KV compression and eviction

H2O (Zhang et al., 2023), SnapKV (Li et al., 2024), FastV (Chen et al., 2024), LOOK-M (Wan et al., 2024), and KVzip (Kim et al., 2025) commit a compressed cache once; Quest (Tang et al., 2024) shows query-aware beats query-agnostic at tight budgets in single-query text — motivation for, but not evidence of, our multi-query claims. SparseVILA (Anonymous, 2025d) is the nearest VLM neighbor for extension E2 (prefill pruning + decode-time retrieval) but prunes irreversibly at prefill and never compares against retrieval pipelines on long-document VQA; Kamera (Anonymous, 2026c) shows the multi-query KV-reuse serving regime is becoming real.

8.3 Trained visual retrieval

ColPali/ColQwen2 (Faysse et al., 2025) is the strong deployable retriever we lose to; M3DocRAG (Cho et al., 2024) and successors are the pipelines CAPS beat. The bar is moving: ColQwen3-class models and small distilled retrievers keep improving, so even a future win over ColQwen2 must be argued as a mechanism/regime result, not a leaderboard entry. A published oracle-analysis on MMLongBench (Anonymous, 2026b) mirrors our fusion finding (best-per-query routing between retrievers gains +0.9; union reranking gains more), and attention-based reranking precedents exist in text (Chen, Gutierrez, and Su, 2025; Wu et al., 2025).

8.4 Benchmarks and the region-level neighborhood

MMLongBench-Doc (Ma et al., 2024) and LongDocURL (Deng et al., 2025) supply the multi-query long-document structure; MMDocIR (Dong et al., 2025) adds layout-level retrieval labels useful for E1's localization sanity checks. The region-level neighborhood is active — RegionRAG (Yuan et al., 2025) (trained retriever-side crops; near-flat on long docs, no long-doc QA), Snappy (Anonymous, 2025c) (training-free patch→OCR-box propagation, localization only), AGREE (Anonymous, 2025a) (reader attention distilled into a trained retriever), VRAG-RL (Wang et al., 2025) and Doc-V* (Anonymous, 2026a) (agentic crop/zoom loops), ViCrop/FOCUS (Zhang et al., 2025; Anonymous, 2025b) (single-image training-free cropping). In our current search we did **not identify prior work in the exact combined cell**: reader-internal attention as the runtime region selector, end-to-end QA at matched token budget vs. page-level retrieval, on long documents, multi-query amortized. This is a provisional literature-map claim, not proof of novelty; it motivates the E1 proposal. Page-selection-then-read itself long predates us (Tito, Karatzas, and Valveny, 2024).

9 Current state, open questions, roadmap

9.1 Scoreboard (2026-07-17)

Item	Status	Where
Multi-query reversibility isolation (page level)	**CONFIRMED** (903 q, both cells)	§7.1
robustness to the KV-cache fix	**VERIFIED** (claim stands)	§4.3
replication at block level	**CONFIRMED** (+0.379, $\text{LB}_{95} = +0.273$)	§7.5.1
Executed retrieval-vs-selection panel at 64K	**DONE** (first of its kind, to our knowledge)	§3, App. A
PAQ v1 vs. ColQwen2-fresh (page level)	**LOST** (−0.099 pooled)	§7.2
Page-level attention selection as a method	**SCOOPED** (CAPS, ACL 2026)	§8.1
Commitment-pressure mechanism	**DEMOTED** (interaction test incl. 0)	§7.3
KV-cache pollution bug	**FOUND, QUANTIFIED, FIXED**	§4.3
PAQ-T recall parity (clean gate)	**CONFIRMED** (0.683/0.626 vs. 0.664/0.536)	§7.4
PAQ-T accuracy read-eval	**EXECUTED = PARITY** (non-inferior, *not* a beat)	§7.4
Lane B: off-benchmark mixture transfer gate	**STOP** (mixture < single head; byproduct L21.H18)	§7.6
PAQ-KV on MMLongBench-Doc (80 docs)	**BEAT HOLDS** (+0.079, $\text{LB}_{95} = +0.041$; audited)	§7.5.1
PAQ-KV vs. gold-page oracle (mmlongdoc)	**MATCHES** (corrected from “exceeds”)	§7.5.1
PAQ-KV on LongDocURL (80 docs)	**HOLD** (+0.019, $\text{LB}_{95} = -0.019$; characterized)	§7.5.2
C-V1 sharp-head block scorer	**NO-GO** (n.s. on longdocurl; hurts mmlongdoc)	§7.6
LongDocURL pure-selection ceilings	**KNIFE-EDGE** (only a hybrid clears comfortably)	§7.5.2
Efficiency table	**PENDING** (unmeasured)	§7.7
Sealed fresh test set (73/45 docs)	**RESERVED, UNTOUCHED** (one-touch final eval)	§9.3
Program scope decision	**HELD** (three-way human fork)	§9.3

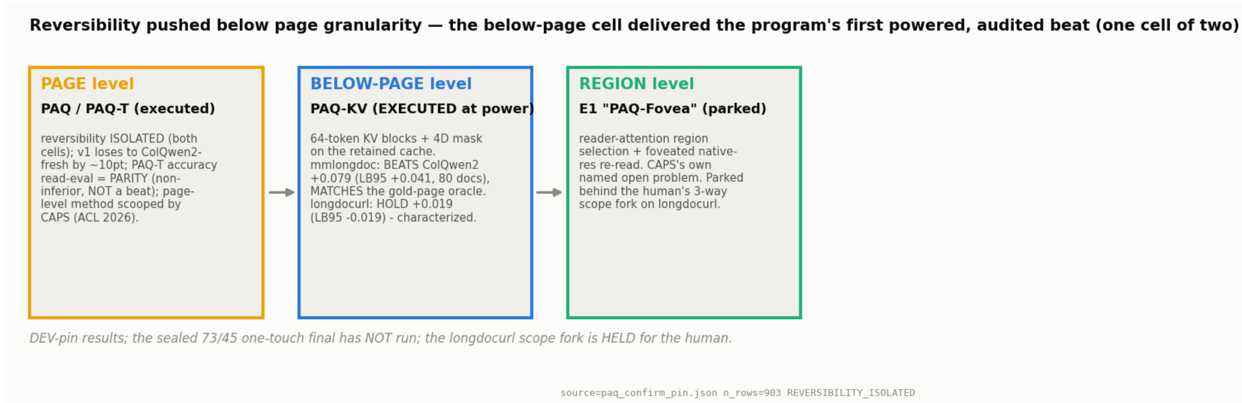

Figure 9: The page → below-page → region granularity map, current state. Page level is executed (v1 lost; PAQ-T parity; the page-level method is CAPS's). The below-page level (PAQ-KV) is now executed at power: a beat on MMLongBench-Doc, a characterized hold on LongDocURL. The region level (E1) is parked behind the human's scope fork.

9.2 The thesis, revised by the evidence

The reframe that survived the earlier review rounds was: **page quantization, not selection quality, is the binding constraint**. The PAQ-KV results at power sharpen it into a two-regime statement (Figure 9):

- On **dispersed-evidence** cells (MMLongBench-Doc: many short facts scattered across pages), the thesis is *vindicated*: below-page granularity beats page-level selection at matched budget (+0.054, $LB_{95} = +0.011$), and the whole package beats the trained retriever while matching the gold-page oracle at a 5% budget (§7.5.1).
- On **conversion-bound** cells (LongDocURL: dense report/table documents), the binding constraint sits *past* selection entirely: ≈41% of queries fail at the native-resolution gold-page oracle, so no selector — page-level, block-level, or oracle — can clear the bar there. Better selection is necessary but not sufficient; the wall is the reader's evidence-to-answer conversion (§7.5.2).

9.3 How the two-lane program resolved, and the held fork

The program mandate is explicit: ship a *method that beats the baselines*; a parity or boundary paper is not an acceptable terminus. The two lanes launched after the PAQ-T parity result have now both resolved:

- **Lane B (off-benchmark selection-adapter): STOPPED** at its cheapest pre-frozen gate — the per-head mixture does not transfer (§7.6); no adapter training was ever funded. The gate did exactly what it was frozen to do.
- **Lane C (PAQ-KV): delivered the program's first powered, audited beat** — on one cell of two (§7.5); its attempted scorer upgrade (C-V1) failed its gate. The lane's pre-computed LongDocURL headwind (full-context − ColQwen2 priced at $LB_{95} = -0.001$ there) materialized as predicted: PAQ-KV reaches full-context-level accuracy on that cell and it is not enough.

The held fork (a human decision, pending). With the mmlongdoc beat standing and the longdocurl characterization in hand, the program is **held** on a three-way scope decision that belongs to the human, not to this document:

1. **Accept the mmlongdoc-scoped result**: submit PAQ-KV as a dispersed-evidence-regime method (a beat on MMLongBench-Doc with a characterized boundary on LongDocURL), and run the sealed one-touch on that scope;

2. **Pursue the non-selection read-resolution lever** on the remaining LongDocURL headroom — de-risked first by a cheap gold-page high-resolution diagnostic; the CPU audit of §7.5.2 predicts near-nil headroom (native resolution already saturates the prefill on 79/80 documents), so this fork expects a fast falsification;

3. **Revisit the target and the reader ceiling**: change the cell mix or the reader rather than continue polishing selection against a conversion wall.

No further GPU work is queued until the fork is decided; overnight autonomous work was stood down deliberately.

The sealed-test discipline (unchanged). The 80-document confirmatory pin has been reused across this program and is therefore **development-only**. The fresh 73/45-document set (73 LongDocURL documents / 433 queries; 45 MMLongBench-Doc documents / 244 queries; zero overlap with the pin) remains **sealed for a single one-touch final evaluation**: ColQwen2 will be run on it once, and the only mandate-grade claim this program will make is a scope-appropriate, adversarially-imputed, document-clustered $\mathrm{LB}_{95} > 0$ on that sealed set. Known weakness, disclosed: the fresh mmlongdoc cell (244 queries / 45 documents) sits below the program's 250-query power floor, so the sealed beat needs the effect to hold at slightly reduced power. *The sealed set has not been evaluated; no number in this document comes from it.*

9.4 Open questions

(1) Does the mmlongdoc beat survive the sealed one-touch — the only mandate-grade evidence still missing? (2) Can the two regimes (dispersed-evidence vs. conversion-bound) be predicted *a priori* for a new cell — the doc-size split and failure taxonomy of §7.5.2 suggest cheap diagnostics exist? (3) Is there a deployable, training-free router that captures part of the measured PAQ-KV∪ColQwen2 complementarity (union ceiling $\mathrm{LB}_{95} = +0.061$) *without* becoming the hybrid the mandate excludes — and should the mandate's scope be revisited instead? (4) Does block-level selection preserve accuracy on multi-hop/cross-page questions at scale (the audit's below-page>page result is one cell; MMLongBench-Doc is 33% cross-page)? (5) The moving bar: ColQwen3-class retrievers may outdate any ColQwen2 comparison by submission time — the claim must stay a regime/mechanism claim. (6) A second VLM (currently everything is a Qwen3-VL-8B case study). (7) The unmeasured efficiency table (§7.7).

10 Limitations

- **Heterogeneity.** Effect sizes differ by cell (e.g. the stale gap is +0.257 vs. +0.163); we never average cells except in the labelled pooled statistic, and per-cell results are primary.

- **Development-pin evidence only.** Every PAQ-KV number in this document comes from the reused 80-document development pin. The sealed 73/45 one-touch final has not run;

until it does, the MMLongBench-Doc beat is development-grade evidence, and we label it so wherever it appears.

- **No efficiency claim.** Page-PAQ re-reads at full resolution; PAQ-KV decodes from the masked retained cache but claims no memory/latency savings (the mask route is the accuracy path; physical gathering is future work); the systems-level story is unmeasured (§7.7).
- **Budget semantics.** The PAQ-KV budget is a *nominal* 5% decode-attention budget (realized ≈5.9–6.0%), with selected blocks' K/V computed under full-document prefill context — the Quest/SnapKV claim semantics, not a matched-token prefill comparison (§4.5).
- **Audit scale and reference provenance.** The mmlongdoc mechanism-audit contrasts are a re-analysis of 30-document development data (the beat itself is 80-document); the LongDocURL full-context reference in Table 7 is the 30-document screen value while its oracle is the cached confirmatory value (disclosed in the caption).
- **Base-model ceiling.** 35.1% of confirmatory queries are oracle-unanswerable (score 0 given exactly the gold pages); only 45.1% are fully oracle-answerable. Absolute numbers are deflated for every arm; oracle is not a strict ceiling (28 PAQ wins over it).
- **Resolution cap.** The high-res vision encoder OOMs on many-page documents, so PAQ-T scoring was capped at a 6,000-token target (and the read-eval's recovery pass used a reduced 4,000-token target on 6 large documents, conservative against PAQ-T) — the resolution lever is under-tested.
- **CAPS overlap.** Expert-head attention selection is CAPS's published method; any PAQ-T claim is limited to the amortized multi-query architecture and must credit CAPS.
- **PAQ-T parity caveats.** The head selection is cross-validated, *benchmark-supervised* model selection (not zero-shot), and the bootstrap does not propagate the CV head-selection uncertainty; the mmlongdoc per-cell $\mathrm{LB}_{95} = +0.001$ is a razor's edge and we do not present it as a per-cell win. Parity, not a beat, is the claim.
- **KV-cache bug residue.** The confirmatory run predates the crop fix (§4.3); the isolation *contrasts* were verified robust, but the absolute PAQ accuracy levels in the confirmatory carry ≈0.04 inflation on the 24-document check and have not been re-run clean at full scale.
- **Weak family ports.** FastV/LOOK-M/KVzip are disclosed proxies and SnapKV is a weak port; we claim nothing against canonical versions of them.
- **Single reader.** All results are a Qwen3-VL-8B case study; generalization to other VLMs is untested.
- **Stale-policy dependence.** The commit-once aggregate uses the first $M = \min(3, k)$ queries in frozen arrival order; order-permutation and M-sensitivity robustness checks were reviewer-requested and remain to be run (the gold-informed static, which is order-free, partially covers this).

Table 8: Every arm, accuracy at b=5%, per cell (verbatim from `research/paq_verdict.json`). The query-blind family sits at/below random gold-recall — the anti-strawman record: beating it is visibly not the claim.

Group	Arm	LongDocURL	MMLongBench-Doc
ours	PAQ	**0.386**	**0.258**
	attn_max (alt-agg)	0.337	0.152
isolation	paq_stale (commit-once)	0.128	0.095
	gold_frequency_static	0.278	0.154
deployable retrieval	ColQwen2-fresh	0.512	0.328
	ColQwen2-stale	0.216	0.138
	CLIP-fresh	0.155	0.164
reference (not competitors)	SnapKV-fresh	0.298	0.157
	full-context	0.546	0.397
	oracle-evidence (not strict)	0.609	0.431
query-blind controls	SnapKV-stale	0.083	0.074
	H2O	0.045	0.039
	FastV (ctx-only proxy)	0.045	0.031
	LOOK-M (→H2O on images)	0.045	0.039
	KVzip (recv_max proxy)	0.047	0.048
	random	0.068	0.022
	truncation	0.033	0.041

A Full executed panel at the gate budget

Secondary statistics (pooled, reported not gated): PAQ beats the best family arm (SnapKV-stale) with min-over-family $\mathrm{LB}_{95} = +0.202$; PAQ − SnapKV-fresh = $+0.094$ $[+0.069, +0.122]$; PAQ − attn_max = $+0.077$ $[+0.048, +0.110]$; PAQ − oracle = -0.199 $[-0.239, -0.161]$.

Gold-page recall at b=5% (per cell, LongDocURL / MMLongBench-Doc): PAQ 0.494/0.443; `gold_frequency_static` 0.381/0.373; `paq_stale` 0.137/0.148; ColQwen2-fresh 0.664/0.536.

B The adversarial review trail (anti-strawman record)

This project's positives were only believed after surviving independent cross-family review; two of them did not survive, and we list the retractions because they are part of the evidence:

1. **Pre-freeze design review** (code-faithfulness + methodology, 2026-07-14): folded best-of-family gating, document-cluster bootstrap, the page-routing estimand disclosure, first-M stale aggregation, and power floors *before* any GPU run.

2. **v1 gate retraction.** The first gate (`PAQ_V1_GO`, PAQ beats the eviction family) *passed and was then retracted*: reviewers + an independent reproduction showed a gold-informed static nearly matched PAQ's recall — the family's weakness was a weak-static artifact, and reversibility had not been isolated. The two-comparator isolation gate (Eq. (6)) replaced it and was frozen before the re-run.

3. **Result review of the isolation positive** (2026-07-14): heterogeneity honestly surfaced (screen-scale MMLongBench static contrast -0.017, CI incl. 0 — later resolved at 80 docs); `attn_best`, a derived arm that peeked at gold task score, was dropped.

4. **Final AAAI-readiness review** (2026-07-15): named the missing deployable retrieval baseline as the #1 threat — running it produced the decisive loss (§7.2); forced the `best_static` → `gold_frequency_static` rename and the LB_{95} relabel; demoted the mechanism claim (§7.3); required the independent-extension and missingness analyses (§7.1).

5. **Efficiency red-team** (2026-07-15): rejected efficiency-only framing (§7.7) and predeclared the non-inferiority bars PAQ-T must meet.

6. **Read-eval BUILD review** (2026-07-16, cross-family): found the **KV-cache pollution bug** (§4.3) as a blocker — the fix, the blast-radius diagnostic, the clean regeneration of the recall gate, and the reversibility robustness check all followed from it; also forced the gate-pinned auditable fold map, per-cell power floors, per-cell missingness sensitivity, and the reframing of the claim as "cross-validated, benchmark-supervised (not zero-shot)."

7. **Read-eval RESULT review** (2026-07-16): one reviewer passed the numbers (hashes, bars, pairing verified); the other **blocked** the complete-case verdict for *informative missingness* (the OOM'd queries were easy for the baseline). The block was resolved by recovering 31/35 missing rows conservatively; the recovery erased the pooled edge (§7.4). This is the program's clearest example of a result review changing a headline.

8. **Lane-B design reviews** (2026-07-16, two rounds: REVISE/HOLD then HOLD): dropped a benchmark-contaminated training source (MoLoRAG), forced fully external selection of every design choice, a fresh held-out document set, 100% coverage with adversarial imputation, packed-context training, an adapter-off invariance test, and honest odds of ∼15–25%.

9. **Combined B+C review and C screen decision** (2026-07-16): imposed the program-level **sealed-test discipline** (the 80-document pin is development-only; the fresh 73/45 set is sealed for one touch); corrected the C-lane spike design (M-RoPE position continuation; the "nominal-5%" budget relabel); and gated the C screen on a decode-identity audit as a kill criterion — which then passed with zero mask-specific argmax changes (§4.5).

10. **C-lane screen and scorer gates executed** (2026-07-16/17): the 30-document screen's pre-frozen gates passed except the mmlongdoc budget-curve monotonicity check (disclosed in §7.5.1); the B-lane transfer gate returned **STOP** (§7.6); the C-V1 scorer gate's three pre-frozen conditions all failed ⇒ **NO-GO**. All three gates did exactly what they were frozen to do.

11. **The mmlongdoc adversarial audit** (2026-07-17): attacked the then-30-document beat on query-dependence, leakage (wrong-question control), beat robustness under the frozen instrument, and the oracle definition. Every mechanistic check passed; one over-claim was **corrected** ("exceeds the gold-page oracle" → "matches"), and the audit's named residual risk — sample size — was retired by the 80-document top-up (§7.5.1).

12. **Stream-D extensions review** (Sol, 2026-07-17; verdict REVISE/BLOCK): corrected the elimination statistics in the LongDocURL extensions note — the symmetric-normal significance bar is only a planning approximation for a percentile bootstrap, and the V0∨V1 union-oracle in fact clears at $LB_{95} = +0.003$ (knife-edge and undeployable, but not "dead") — and classified the PAQ-KV∪ColQwen2 router as a hybrid-system result that cannot satisfy the standalone-beat mandate. Both corrections are folded into §7.5.2.

C Reproducibility

- **Harness** (frozen): `scripts/paq.py` (v1 sha256 `735eecaa…`, commit `ca66292`; v2 `e9e06aad…`, commit `f7e7a53`; v3 = v2 + confirmatory pin, re-scored at commit `b653e85`). Retrieval arms: `scripts/paq_retrieval_arms.p` (ColQwen2 `vidore/colqwen2-v1.0` + CLIP; incremental merge — the frozen gate math and the retrieval-free rows are untouched). PAQ-T gate: `scripts/paq_t_gate.py`. Figures: `scripts/paq_figs.py` (all numbers read from the verdict JSONs; every figure carries a source stamp `pin`·`n_rows`·`verdict`).

- **Pins**: screen `research/paq_pinned_subset.json` (sha256 `b194ed84…`, chmod 444, byte-verified by the harness); confirmatory `research/paq_confirm_pin.json` (80 docs/cell; the screen pin is its prefix).

- **Verdicts**: `research/paq_verdict.json` (gate + budget curves + retrieval secondary + strata), `research/paq_t_gate_verdict.json` (clean, crop-fixed); per-query per-arm rows in `research/paq_results.jsonl`; cached retrieval scores in `research/paq_retrieval_scores.json`.

- **Read-eval + integrity artifacts (2026-07-16)**: harness `scripts/paq_t_readeval.py` with pre-registered bars in `research/paq_t_readeval_frozen_meta.json` (frozen, sha-pinned, before the run); verdicts `research/paq_t_readeval_verdict.json` (complete-case) and `research/paq_t_readeval_combined_verdi` (near-full coverage; the headline); recovery `scripts/paq_t_readeval_recover.py`. KV-bug artifacts: blast radius `research/paq_kvcrop_diag.json`, reversibility robustness `research/paq_revcrop_verdict.json` (`scripts/paq_reversibility_crop_check.py`).

- **Lane artifacts (2026-07-16)**: C-lane design `research/design_notes/paqc_e2_paqkv_design_2026-07-16.md`, spike `scripts/paqc_spike.py` with `research/paqc_spike_test{A,B}.json` and `research/paqc_identity_audit.j` B-lane design `research/design_notes/paqt_lora_calibration_design_2026-07-16.md` (incl. the folded review addenda) and data/harness spec `research/design_notes/paqb_data_harness_spec_2026-07-16.md` (`scripts/paqb_prep.py`; sealed fresh pin + provenance index). Review archive: `research/reviews/paqt_2026-07-16`

- **PAQ-KV at-power + lane-verdict artifacts (2026-07-17)**: screen `research/paqc_screen_verdict.json` (with frozen meta `research/paqc_screen_frozen_meta.json`); top-ups `research/paqc_topup_verdict.json` (LongDocURL, `HOLD`) and `research/paqc_mmlo_topup_verdict.json` (MMLongBench-Doc, `BEAT_HOLDS`); adversarial audit `research/paqc_mmlongdoc_audit_verdict.json`; C-V1 scorer gate `research/paqc_scorer_verdic` (bar frozen in `research/paqc_scorer_bar_frozen_meta.json`); B-lane transfer gate `research/paqb_transfer_gate.` (`STOP`); LongDocURL characterization `research/design_notes/paqc_longdocurl_extensions_2026-07-17.md` with cross-family review `research/reviews/paqc_streamD_2026-07-17/sol_verdict.txt`. Figures `fig_h`–`fig_k` are generated from these JSONs by `scripts/paq_figs.py` (light + dark; nothing hand-entered).

- **Gate + bootstrap + pairing** are unit-tested (`paq.py --selftest`); the belief gate, arm classifications, and strata were pre-registered in the frozen bar.

- **Commands**: gate `python scripts/paq.py --run --cell vision --gate-only`; full contest `--run --cell all`; scoring `--score`; figures `python scripts/paq_figs.py`.

D Timeout recovery note

The confirmatory hit its 10-hour wall-clock limit during final scoring (exit 124) at 903/907 vision queries. Because the harness flushes each row as it completes, the run was recovered by re-scoring the flushed rows on CPU (no GPU re-run, no selection re-computation); the 4 unscored queries are the `mi_phone` rows of §7.1, and the worst-case imputation leaves every gate conclusion unchanged. A separate single-query text cell ("counting") was truncated by the same timeout and is out of scope for the multi-query claims; it was not re-run.

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